


Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

Discrete Fourier Transform

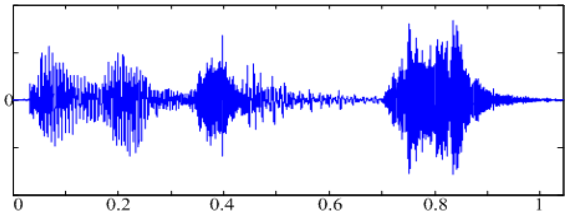
- Definition of the Discrete Fourier Transform
- Properties of the Discrete Fourier Transform
- Calculating Linear Convolution using DFT
- Analyzing Signal Spectra using DFT
- Calculating DFT using MATLAB

Utilizing DFT to analyze the signal spectrum

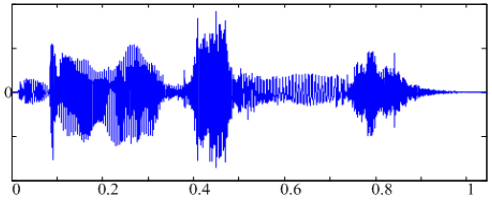
- DFT analysis of the frequency spectrum of a continuous non-periodic signal
- Aliasing, Leakage, and the Fence Effect
- Parameter Selection for Signal Spectrum Analysis by DFT
- Applications of Signal Spectrum Analysis Using DFT

DFT Analysis of the Spectrum of Continuous Non-Periodic Signals

Female voice time-domain waveform



Male voice time-domain waveform



$$x(t) \xrightarrow{\text{Discretization}} x[k] \xrightarrow{\text{Periodization}} \tilde{x}[k] \xrightarrow{\text{Analyze a period}} x[k] \xrightarrow{\text{DFT}}$$

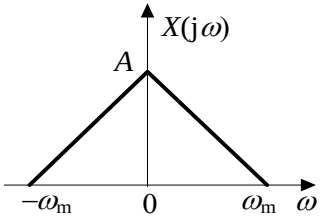
$$X(\mathrm{j}\omega) \xrightarrow{\text{Periodization}} X(\mathrm{e}^{\mathrm{j}\Omega}) \xrightarrow{\text{Discretization}} \tilde{X}[m] \xrightarrow{\text{Analyze a period}} X[m]$$

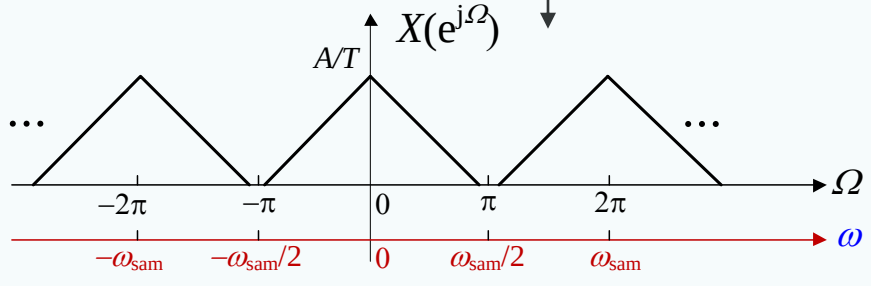
DFT Analysis of the Spectrum of Continuous Non-Periodic Signals

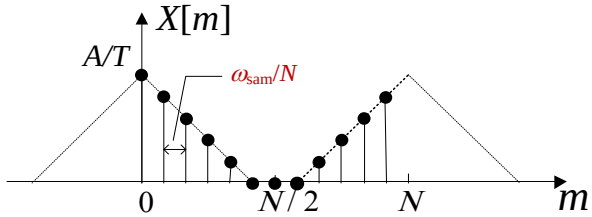
$x(t)$ → Sampling → $x[k]$ → DFT → $X[m]$

$$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X[j(\omega + n\omega_{sam})]$$

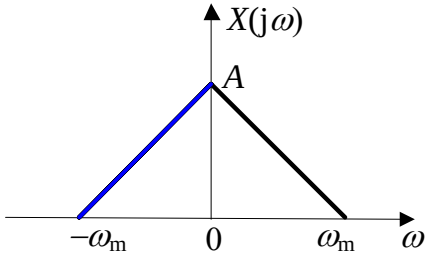
$$X[m] = X(e^{j\Omega})\Big|_{\Omega=\frac{2\pi}{N}m}, m = 0, \cdots, N-1$$

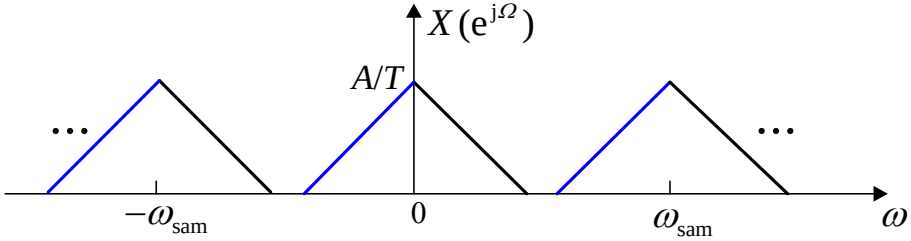






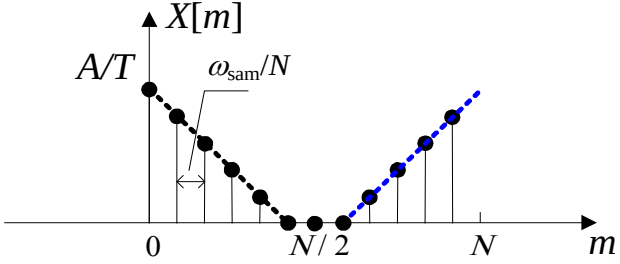
DFT Analysis of the Spectrum of Continuous Non-Periodic Signals





$$\omega = \frac{\omega_{\text{sam}}}{N} m \qquad 0 \le m \le \frac{N}{2}$$

$$\omega = \frac{\omega_{\text{sam}}}{N} m - \omega_{\text{sam}} \qquad \frac{N}{2} < m \le N - 1$$



$$N : \omega_{\text{sam}} = m : \omega$$

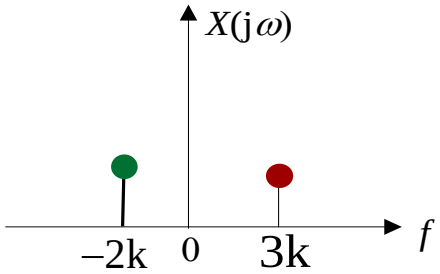
Example:It is known that the highest frequency of the speech signal $x(t)$ is f_m=3.4kHz, Sample $x(t)$ using f_{sam}=8kHz . If a DFT of 1600 points is performed on the sampled signal, determine the corresponding continuous frequency spectrum points f_1 and f_2 in $X[m]$ for m=600 and m=1200.

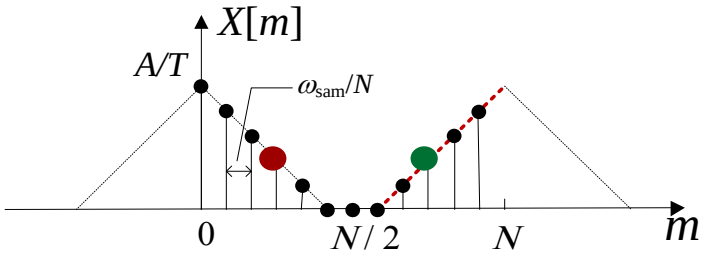
Solution: m=600 $m \le N/2$

$$f_1 = \frac{f_{sam}}{N} m = \frac{8}{1600} \times 600\text{kHz} = 3\text{kHz}$$

m=1200 $N/2 < m \le N-1$

$$f_2 = \frac{f_{sam}}{N} (m - N) = \frac{8}{1600} \times (-400)\text{kHz} = -2\text{kHz}$$



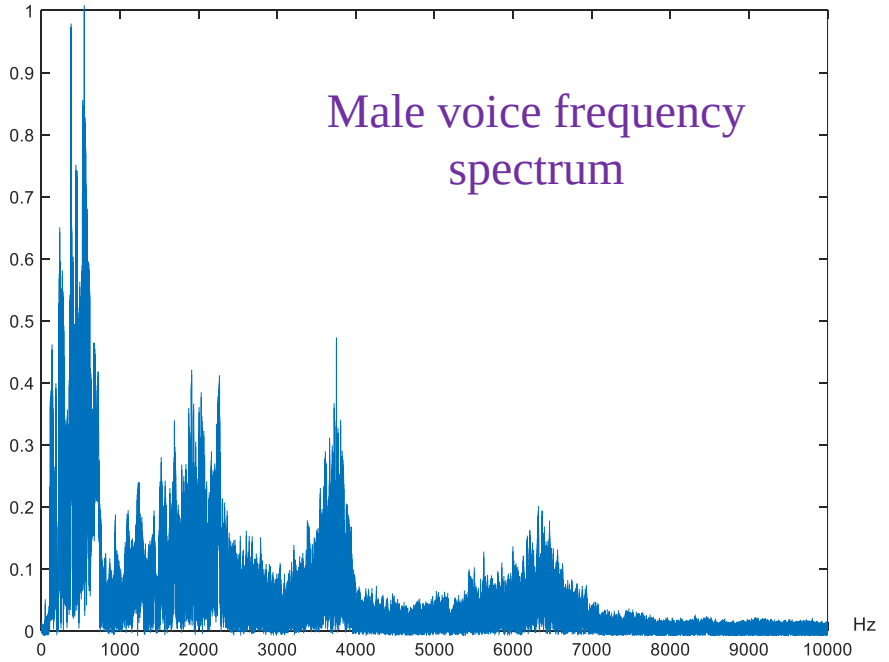


Example: Try to calculate the spectrum of the sound signal using MATLAB.

MATLAB Program as follows:

```
[x,fs]=audioread(' Male voice.wav');
ws = 2*pi*fs;
N=length(x);
X=fftshift(fft(x(:,1),N));
w=-ws/2+(0:N-1)*ws/N;
figure,plot(w/(2*pi),abs(X)/max(abs(X)));
xlim([0,10000]);
```

The execution results are as follows:



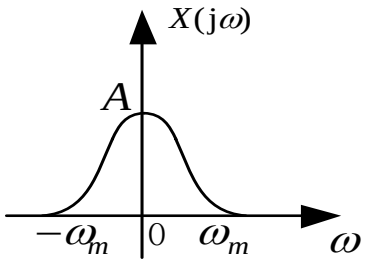
Utilizing DFT to analyze the signal spectrum

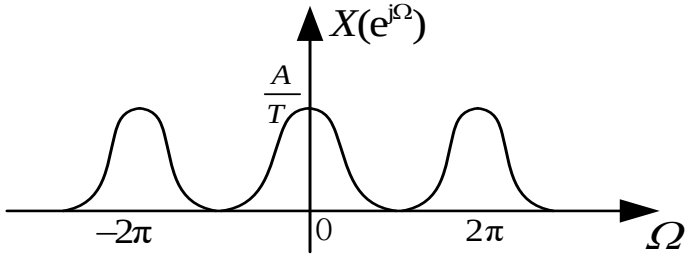
$x(t)$ —Sampling→ $x[k]$ —DFT→ $X[m]$

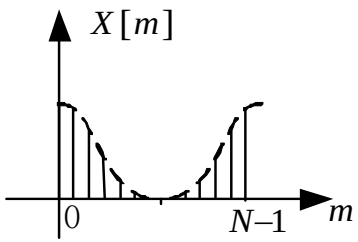
Time is limited
Bandwidth is limited

$$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X[j(\omega + n\omega_{sam})]$$

$$X[m] = X(e^{j\Omega})\Big|_{\Omega=\frac{2\pi}{N}m}, m = 0, \cdots, N-1$$







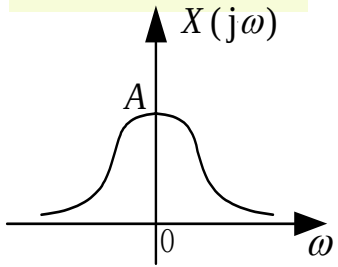
Perfect !

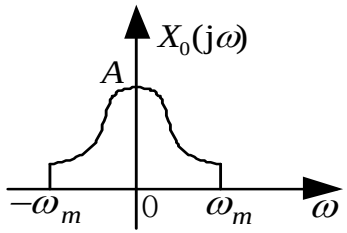
The sampled spectrum is free from aliasing, $x[k]$ is of finite length.

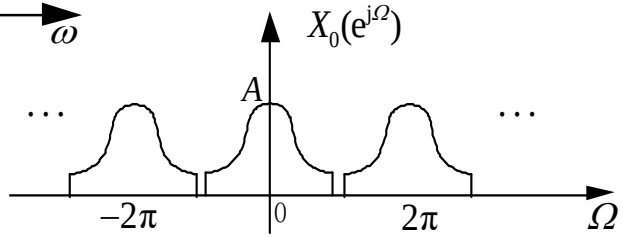
Aliasing Phenomenon

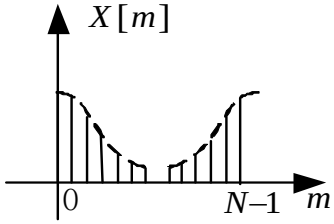
$x(t)$ —Anti-aliasing→ $x_0(t)$ —Sampling→ $x[k]$ —DFT→ $X[m]$

Time is limited
Bandwidth is infinite



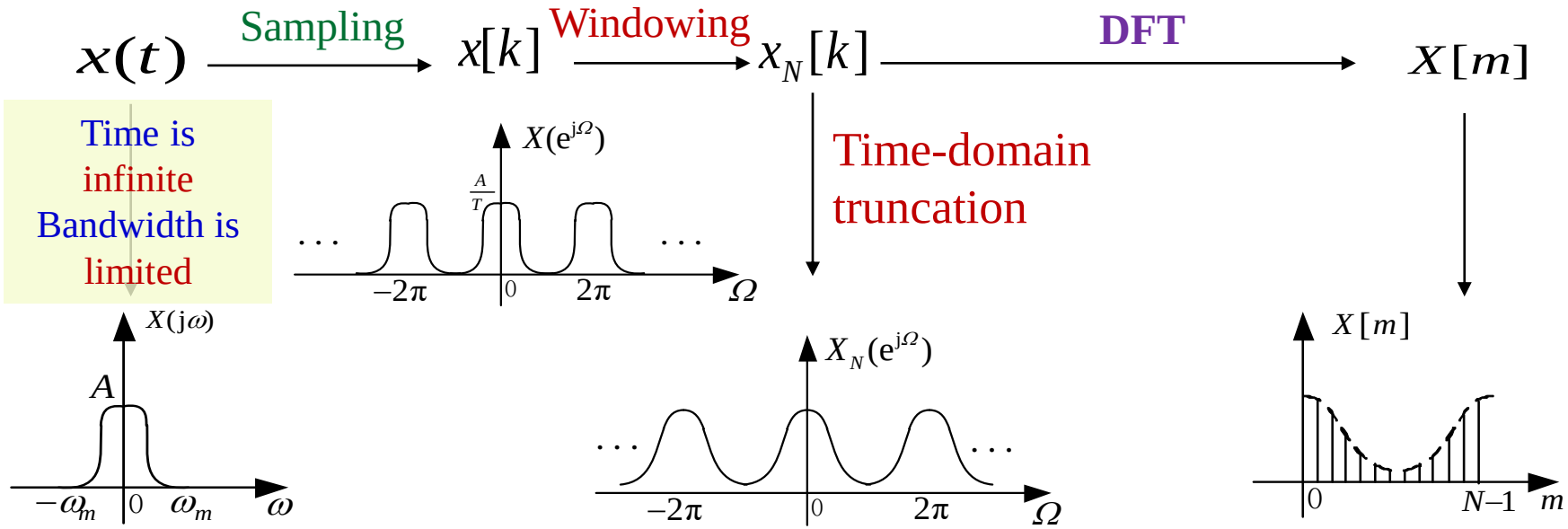






Using DFT to analyze a continuous-time signal that is time-limited but has an infinite bandwidth can result in aliasing phenomena. Methods to improve aliasing phenomena: Anti-alias filtering, reducing the sampling interval T.

Leakage Phenomenon



Using DFT to analyze a continuous-time signal that is time-infinite but has a finite bandwidth can result in leakage phenomena Methods to improve leakage phenomena: Choosing the appropriate window function and width

DFT Analysis of the Spectrum of Non-Periodic Signals

- For an infinitely long band-limited continuous-time signal, what is the process of using DFT to analyze its spectrum?

Sampling—Windowing—DFT—Establish the relationship between $X[m]$ and $X(\mathrm{j}\omega)$

- What is the process of using DFT to analyze the spectrum of an infinitely long discrete sequence?

Windowing—DFT—Establish the relationship between $X[m]$ and $X(\mathrm{e}^{\mathrm{j}\Omega})$

- What is the process of using DFT to analyze the spectrum of a finite-length discrete sequence?

DFT—Establish the relationship between $X[m]$ and $X(\mathrm{e}^{\mathrm{j}\Omega})$

Leakage Phenomenon

$$x(t) \xrightarrow{\text{Sampling}} x[k] \xrightarrow{\text{Windowing}} x_N[k] \xrightarrow{\text{DFT}} X[m]$$

$$x_N[k] = x[k] \cdot w_N[k]$$

$$X_N(\mathrm{e}^{\mathrm{j}\Omega}) = X(\mathrm{e}^{\mathrm{j}\Omega}) * W_N(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\mathrm{e}^{\mathrm{j}\theta}) W_N(\mathrm{e}^{\mathrm{j}(\Omega-\theta)})\mathrm{d}\theta$$

Rectangular window

$$w_N[k] = R_N[k] = \begin{cases} 1 & 0 \le k \le N-1 \\ 0 & \text{others} \end{cases}$$

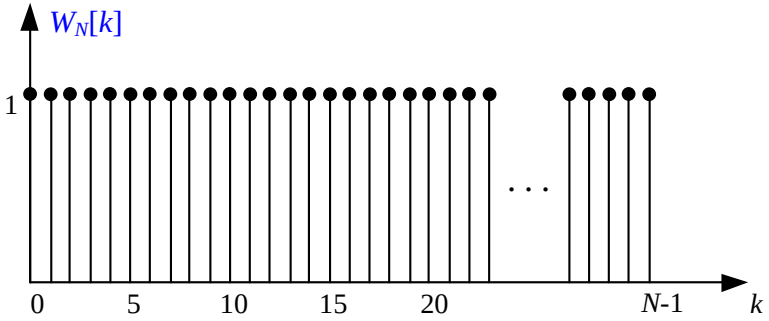
$$W_N(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{\sin(N\Omega/2)}{\sin(\Omega/2)} \mathrm{e}^{-\mathrm{j}\frac{N-1}{2}\Omega}$$

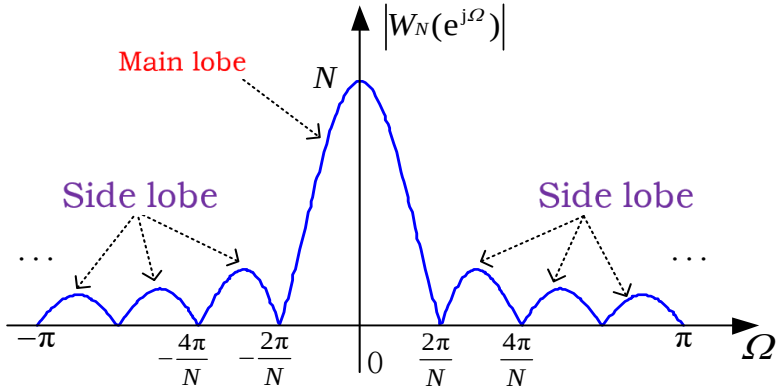
Leakage Phenomenon

Rectangular window

$$w_N[k] = R_N[k] = \begin{cases} 1 & 0 \le k \le N-1 \\ 0 & \text{others} \end{cases}$$

$$W_N(e^{j\Omega}) = \frac{\sin(N\Omega/2)}{\sin(\Omega/2)} e^{-j\Omega\frac{N-1}{2}}$$



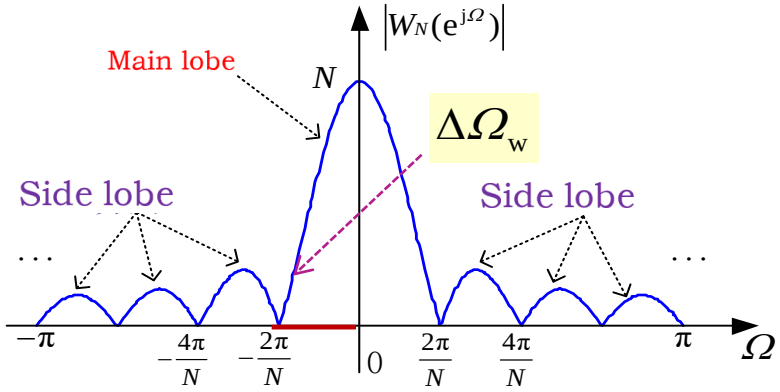


The spectrum of the rectangular window function consists of the main lobe and side lobes, with the main lobe peaking at the origin. Due to the abrupt truncation of the rectangular window function at its two endpoints, there are some high-amplitude high-frequency side lobe components in the spectrum.

Leakage Phenomenon

$$W_N(e^{j\Omega}) = \frac{\sin(N\Omega/2)}{\sin(\Omega/2)} e^{-j\Omega\frac{N-1}{2}}$$

$$\left|\frac{W(e^{j3\pi/N})}{W(e^{j0})}\right| = \left|\frac{\sin(3\pi/2)}{N\sin(3\pi/2N)}\right| \approx \frac{2}{3\pi}$$

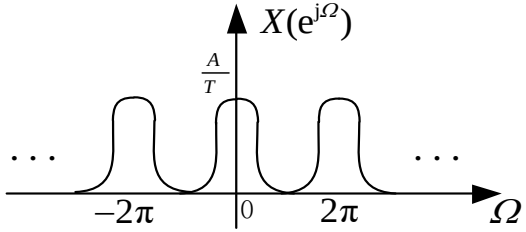


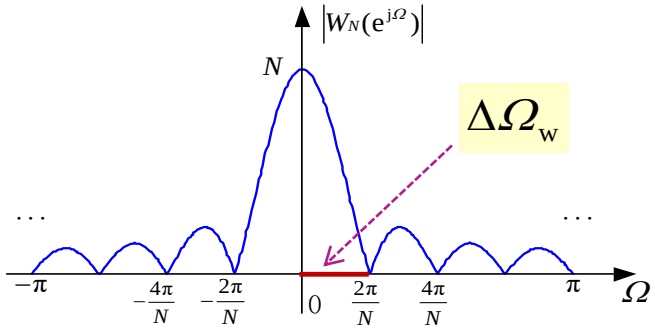
Side lobe peak decay:

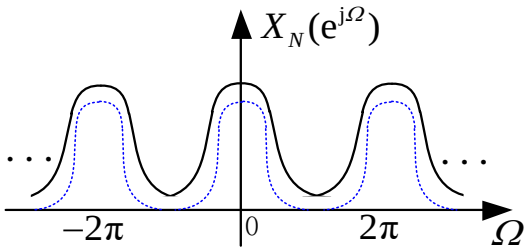
$$A = -20\lg\left|\frac{W(e^{j3\pi/N})}{W(e^{j0})}\right| \approx 13.46 \text{ dB}$$

Main lobe effective width: $\Delta\Omega_w = \frac{2\pi}{N}$

Leakage Phenomenon







Main lobe effective width: $\Delta\Omega_w = \frac{2\pi}{N}$

$$X_N(e^{j\Omega}) = X(e^{j\Omega}) * W_N(e^{j\Omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta}) W_N(e^{j(\Omega-\theta)}) d\theta$$

Example: Analyze the spectrum of an infinitely long cosine signal

$$x(t)=\cos(\omega_0 t),\quad -\infty<t<\infty$$

Solution:

$$X(\mathrm{j}\omega)=\pi[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)]$$

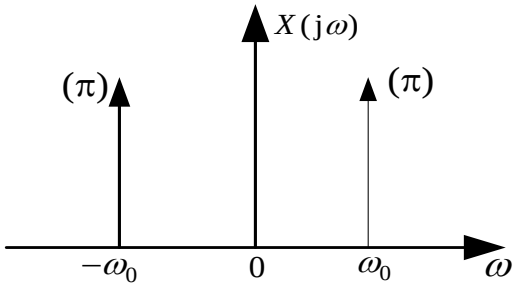
$$x[k]=\cos(\omega_0 kT)=\cos(\Omega_0 k)$$

$$X(\mathrm{e}^{\mathrm{j}\Omega})=\pi\cdot[\delta_{2\pi}(\Omega+\Omega_0)+\delta_{2\pi}(\Omega-\Omega_0)]$$

$$x_N[k]=x[k]\cdot w_N[k]$$

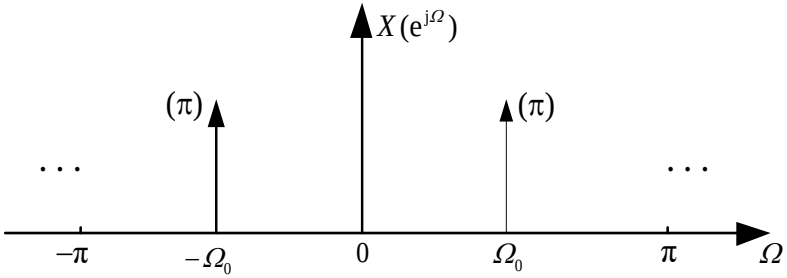
$$X_N(\mathrm{e}^{\mathrm{j}\Omega})=X(\mathrm{e}^{\mathrm{j}\Omega})*W_N(\mathrm{e}^{\mathrm{j}\Omega})$$

$$X_N(\mathrm{e}^{\mathrm{j}\Omega})=0.5\cdot[W_N(\Omega+\Omega_0)+W_N(\Omega-\Omega_0)]$$



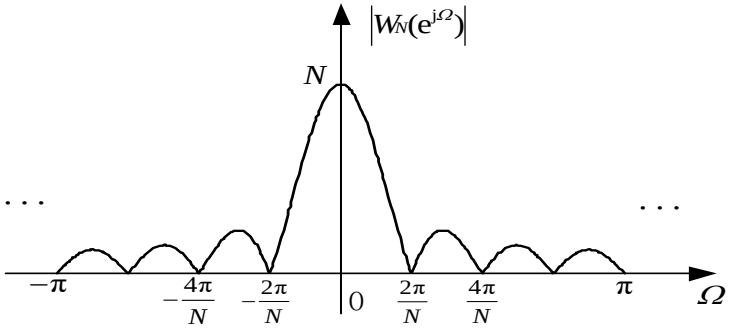
Example: Analyze the spectrum of an infinitely long cosine signal

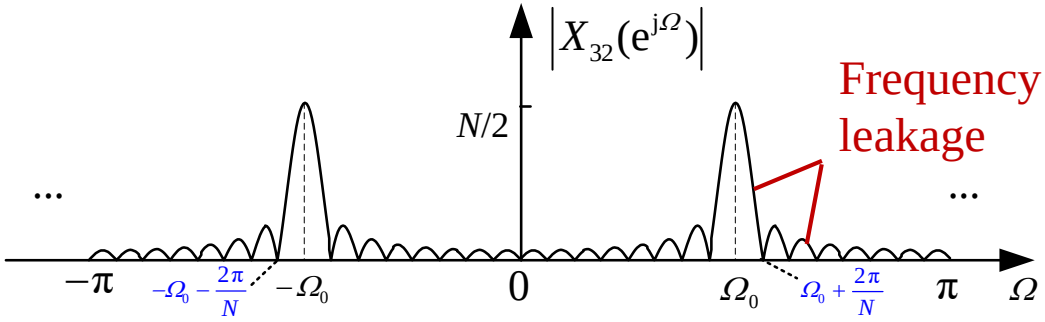
$$x(t)=\cos(\omega_0 t),\quad -\infty<t<\infty$$



$$X(e^{j\Omega})=\pi\cdot[\delta_{2\pi}(\Omega+\Omega_0)+\delta_{2\pi}(\Omega-\Omega_0)]$$

$$X_N(e^{j\Omega})=0.5\cdot[W_N(\Omega+\Omega_0)+W_N(\Omega-\Omega_0)]$$

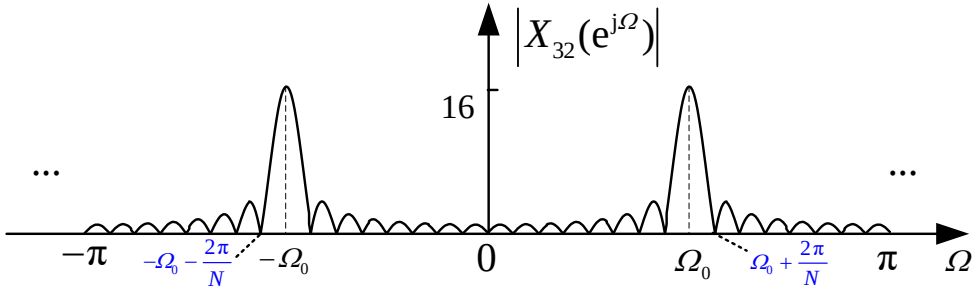




Example: Analyze the spectrum of an infinitely long cosine signal

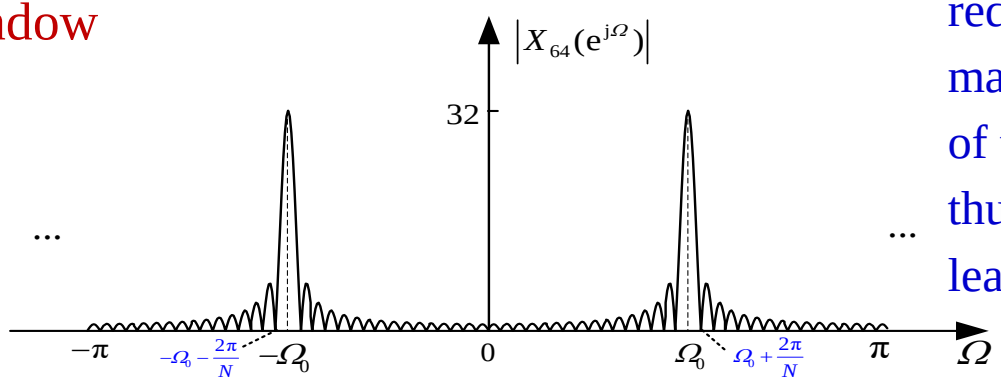
$$x(t)=\cos(\omega_0 t),\quad -\infty<t<\infty$$

N=32



Change the window width N

N=64

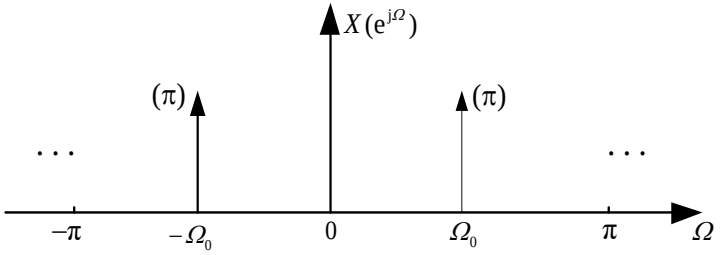


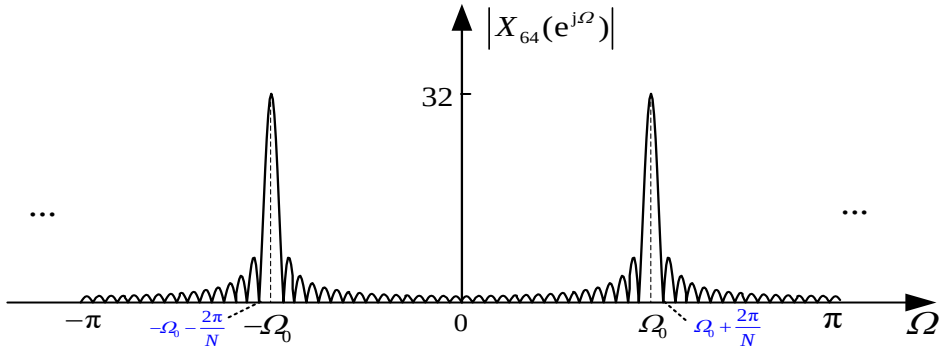
Increasing N can reduce the width of the main lobe and side lobes of the window function, thus reducing frequency leakage.

Increasing N does not improve side lobe decay

Example: Analyze the spectrum of an infinitely long cosine signal

$$x(t)=\cos(\omega_0 t),\quad -\infty<t<\infty$$





Due to windowing in the time domain, frequency leakage occurs, causing a single spectral line to become a spectral peak with a certain width.

If the signal contains two cosine signals of different frequencies, how should N be chosen so that the two spectral peaks can be distinguished?

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=120\text{Hz}$

If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Solution:

$$x[k]=x(t)\Big|_{t=kT}=\cos(2\pi f_1 k/f_{sam})+\cos(2\pi f_2 k/f_{sam})$$

$$=\cos(\frac{\pi}{3}k)+\cos(\frac{6\pi}{15}k)$$

$$x_N[k]=x[k]\cdot w_N[k]$$

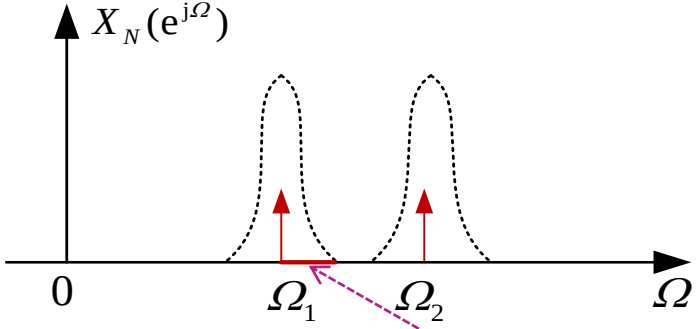
-----The signal $x[k]$ is multiplied by a rectangular window

$$X_N(e^{j\Omega})=0.5\cdot[W_N(\Omega+\Omega_1)+W_N(\Omega-\Omega_1)]+0.5\cdot[W_N(\Omega+\Omega_2)+W_N(\Omega-\Omega_2)]$$

Window Width Selection

$$X_N(e^{j\Omega}) = 0.5\cdot[W_N(\Omega+\Omega_1)+W_N(\Omega-\Omega_1)]+0.5\cdot[W_N(\Omega+\Omega_2)+W_N(\Omega-\Omega_2)]$$

Rectangular window.



The main lobe effective width of a rectangular window

$$\Delta\Omega_w = \frac{2\pi}{N}$$

$$\Delta\Omega = \Omega_2 - \Omega_1 = \frac{2\pi(f_2 - f_1)}{f_{sam}} = \frac{2\pi\Delta f}{f_{sam}}$$

$$\Delta\Omega_w = \Delta\Omega/2$$

$$\Delta\Omega/2 \geq \Delta\Omega_w$$

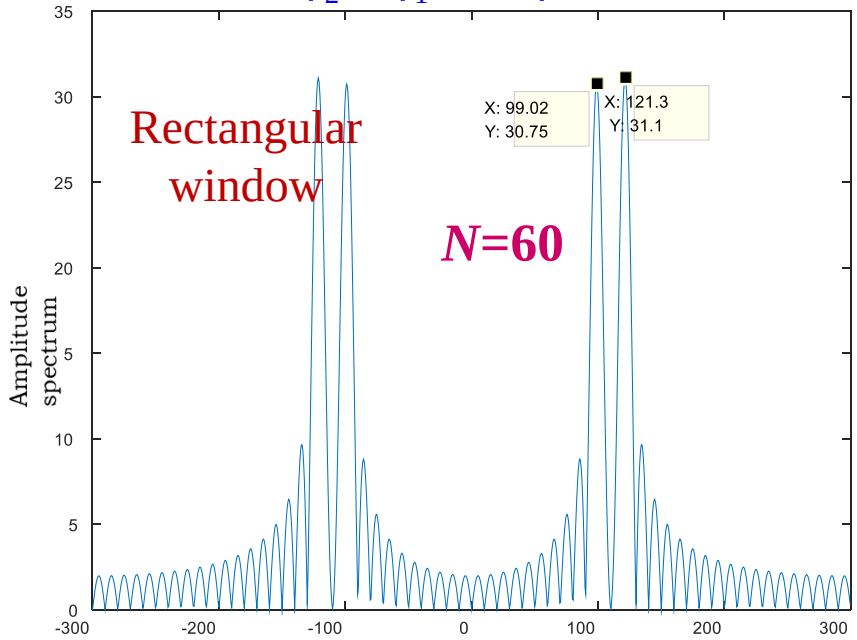
$$\frac{\pi\Delta f}{f_{sam}} \geq \frac{2\pi}{N}$$

Width of the rectangular window

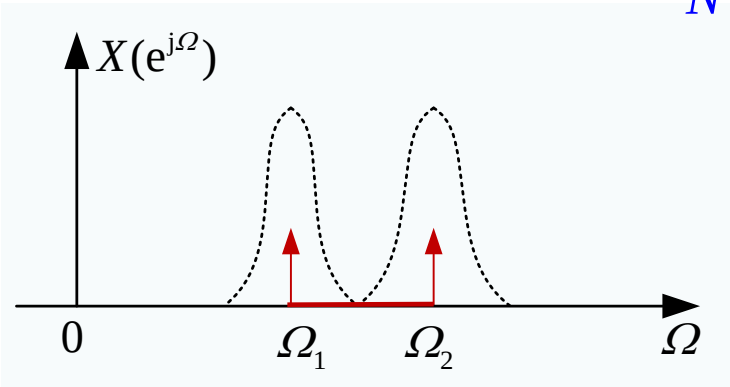
$$N \geq \frac{2f_{sam}}{\Delta f} = \frac{2f_{sam}}{f_2 - f_1}$$

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=120\text{Hz}$ If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Solution: $$N=\frac{2f_{sam}}{f_2-f_1}=\frac{2f_{sam}}{\Delta f}=\frac{2\times 600}{20}=60$$



The main lobe effective width of a rectangular window $$\Delta\Omega_w=\frac{2\pi}{N}$$



$$\Delta\Omega=2\Delta\Omega_w$$

$$N=\frac{2f_{sam}}{\Delta f}=\frac{2f_{sam}}{f_2-f_1}$$

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100Hz$, $f_2=120Hz$. If the signal is sampled at a sampling frequency of $f_{sam}=600Hz$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Solution：

The MATLAB program is as follows:

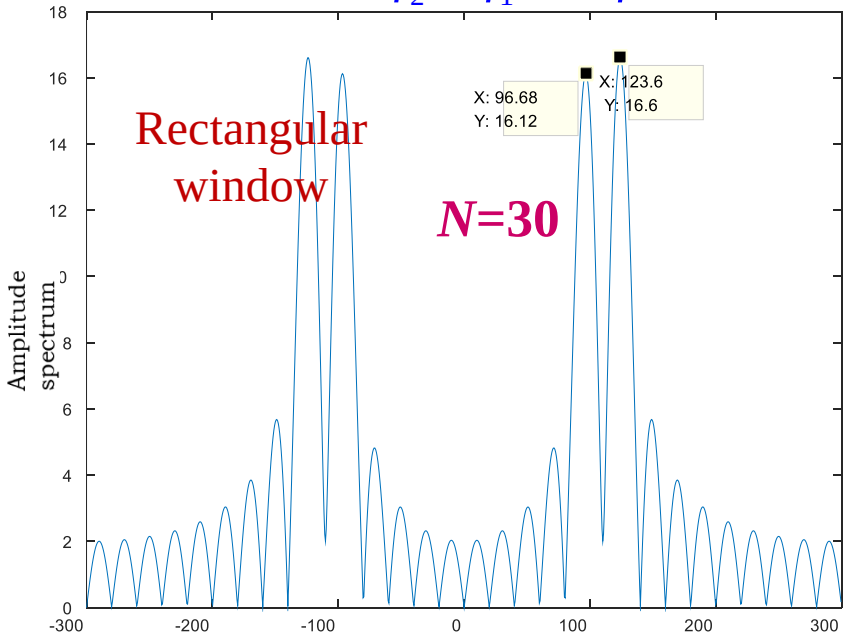
```
N=60;
L=1024;
f1=100;  f2=120;
fs=600;
T=1/fs;
ws=2*pi*fs;
```

Using a rectangular window

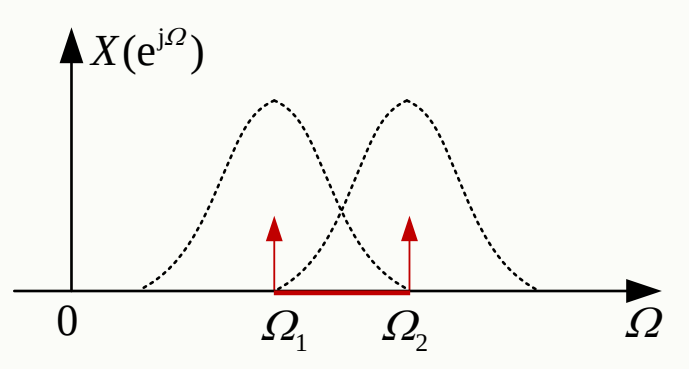
```
t=(0:N-1)*T;
x=cos(2*pi*f1*t)+cos(2*pi*f2*t);
X=fftshift(fft(x,L));
w=(-ws/2+(0:L-1)*ws/L)/(2*pi);
plot(w,abs(X));
ylabel('Amplitude spectrum');
```

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=120\text{Hz}$ If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Solution: $N=\dfrac{f_{sam}}{f_2-f_1}=\dfrac{f_{sam}}{\Delta f}=\dfrac{600}{20}=30$



The main lobe effective width of a rectangular window $\Delta\Omega_w=\dfrac{2\pi}{N}$

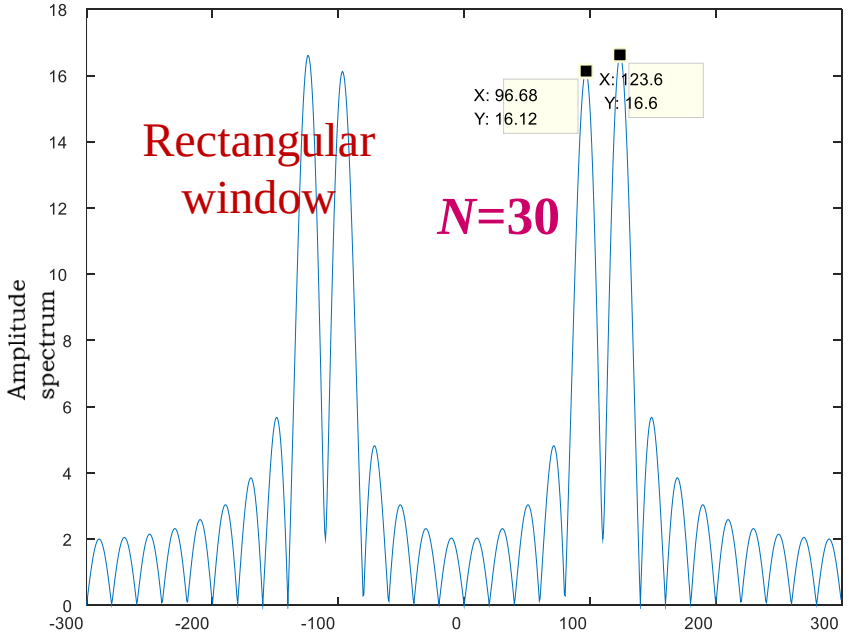


$$\Delta\Omega_w=\Delta\Omega$$

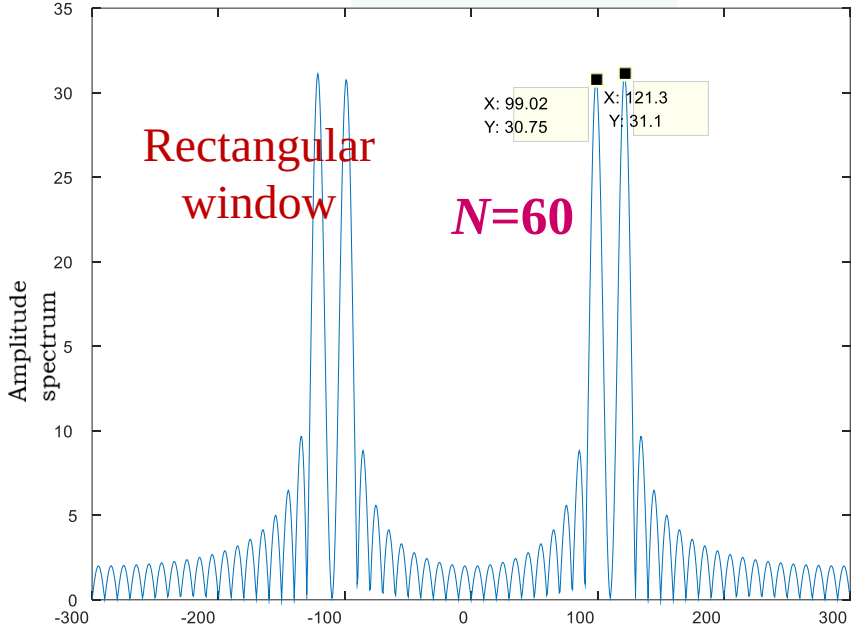
$$N=\frac{f_{sam}}{\Delta f}=\frac{f_{sam}}{f_2-f_1}$$

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=120\text{Hz}$ If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Solution: $\Delta\Omega_w = \Delta\Omega$



$\Delta\Omega_w = \Delta\Omega/2$



Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=120\text{Hz}$ If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

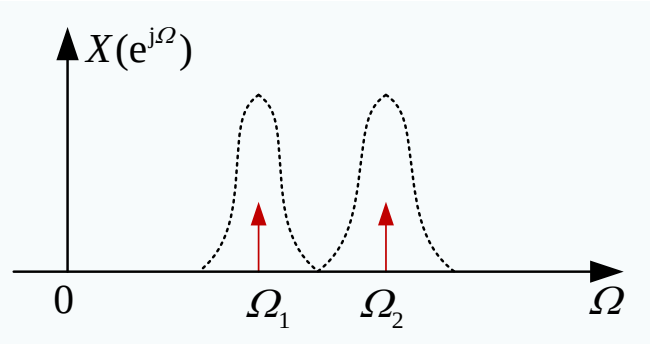
When the amplitudes of two frequency points are essentially the same, to ensure that the two spectral peaks can be distinguished, the selection of the window width N is:

$$\Delta\Omega = 2\Delta\Omega_w$$

The effective width of the main lobe of a rectangular window

$$\Delta\Omega_w = \frac{2\pi}{N}$$

$$N = \frac{2f_{sam}}{\Delta f} = \frac{2f_{sam}}{f_2 - f_1}$$



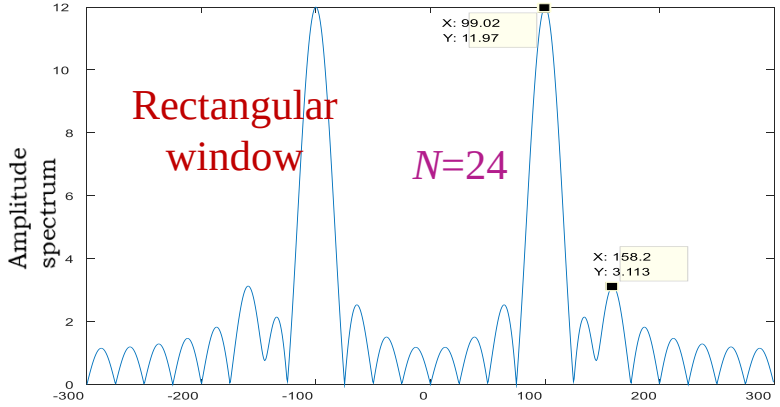
If the amplitude values of the two frequency points differ significantly, how to choose N?

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+0.15\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=150\text{Hz}$ If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$, calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

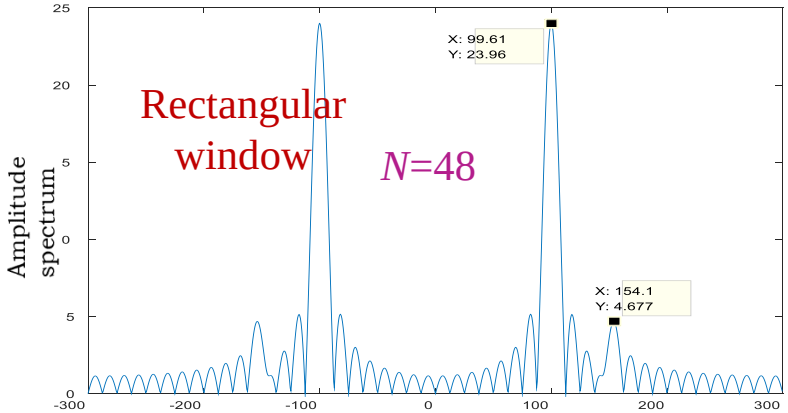
Solution: $x[k]=x(t)\Big|_{t=kT}=\cos(2\pi f_1 k/f_{sam})+0.15\cos(2\pi f_2 k/f_{sam})=\cos(\frac{\pi}{3}k)+0.15\cos(\frac{\pi}{4}k)$

$\Delta\Omega_w=\Delta\Omega/2$

$$N=\frac{2f_{sam}}{f_2-f_1}=\frac{2f_{sam}}{\Delta f}=\frac{2\times 600}{50}=24$$



$\Delta\Omega_w=\Delta\Omega/2$

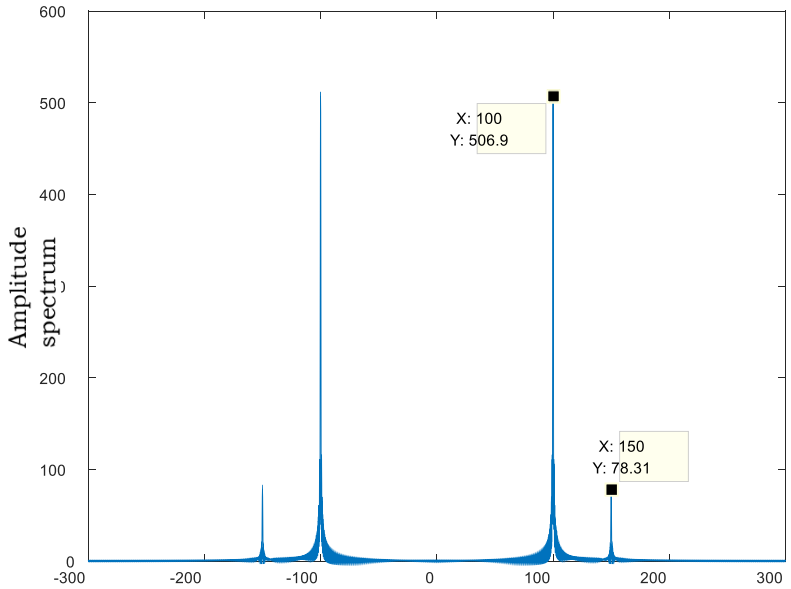


Increase N

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+0.15\cos(2\pi f_2 t)$, $f_1=100$Hz, $f_2=150$Hz If the signal is sampled at a sampling frequency of $f_{sam}=600$Hz , calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Rectangular window

$N=1024$



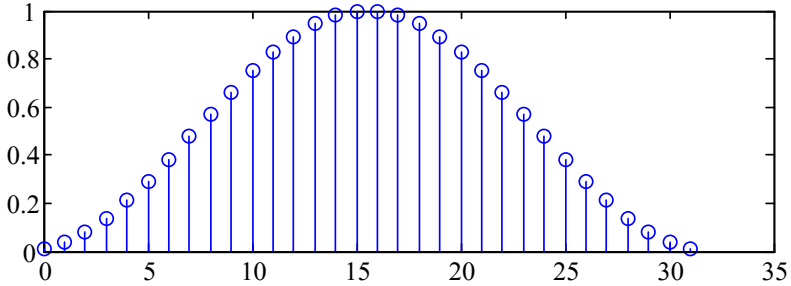
When the amplitude values of two frequency points differ significantly, and a rectangular window is used for windowing, in order to distinguish between the two spectral peaks, the value of N needs to be quite large.

The ratio of the amplitude values of the two frequency points is approximately equal to 1/0.15, and both the frequency position and amplitude value estimations are correct.

Choose a window with greater side lobe attenuation.

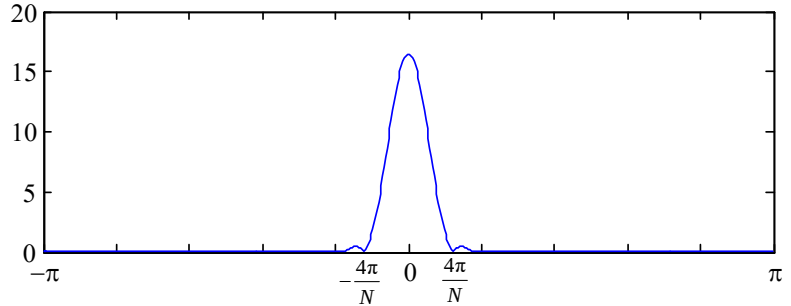
Leakage Phenomenon

Hanning $w_N[k]=[\frac{1}{2}-\frac{1}{2}\cos(\frac{2\pi}{N-1}k)]R_N[k]=\begin{cases}0.5-0.5\cos(2\pi k/(N-1)) & 0\le k\le N-1\\ 0 & \text{others}\end{cases}$



Main lobe effective width: $\Delta\Omega_w=\frac{4\pi}{N}$

Side lobe peak decay: $A\approx 31\ \text{dB}$



Leakage Phenomenon

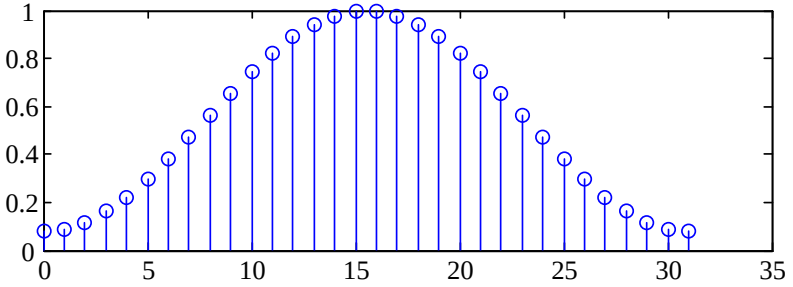
Hamming $w_N[k]=[0.54-0.46\cos(\frac{2\pi}{N-1}k)]R_N[k]=\begin{cases}0.54-0.46\cos(2\pi k/(N-1)) & 0\le k\le N-1\\ 0 & \text{others}\end{cases}$

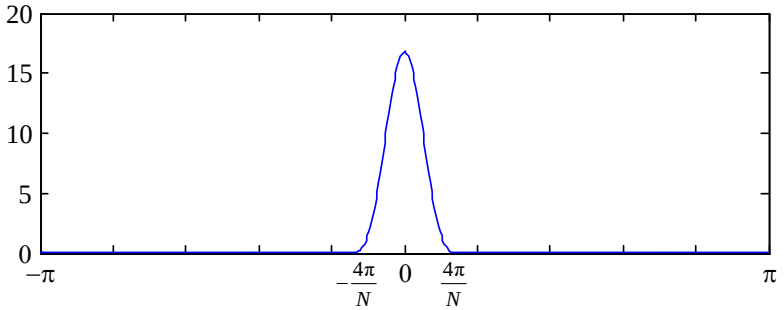
Main lobe effective width:

$$\Delta\Omega_w=\frac{4\pi}{N}$$

Side lobe peak decay:

$$A\approx 41\ \text{dB}$$

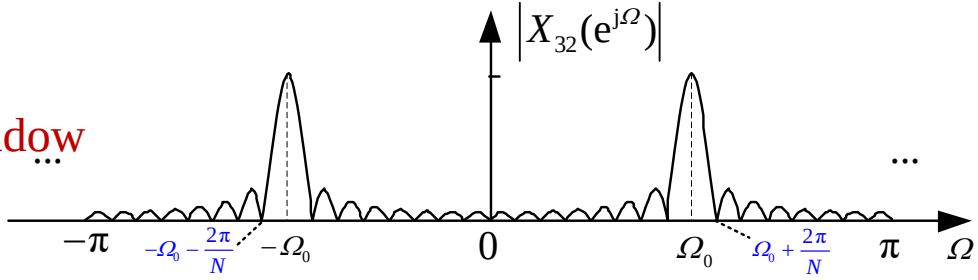




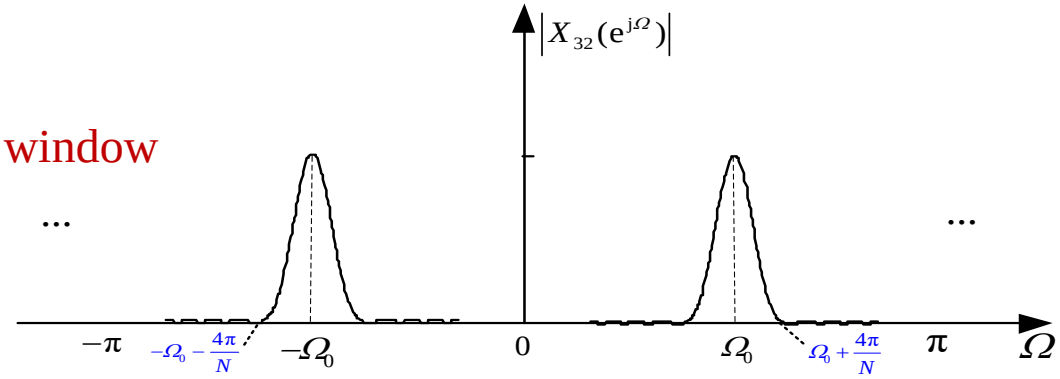
Example: Analyze the spectrum of an infinitely long cosine signal

$$x(t)=\cos(\omega_0 t),\quad -\infty<t<\infty$$

Rectangular window



Hamming window



Change the type of window

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+0.15\cos(2\pi f_2 t)$, $f_1=100$Hz, $f_2=150$Hz If the signal is sampled at a sampling frequency of $f_{sam}=600$Hz , calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis

Solution：

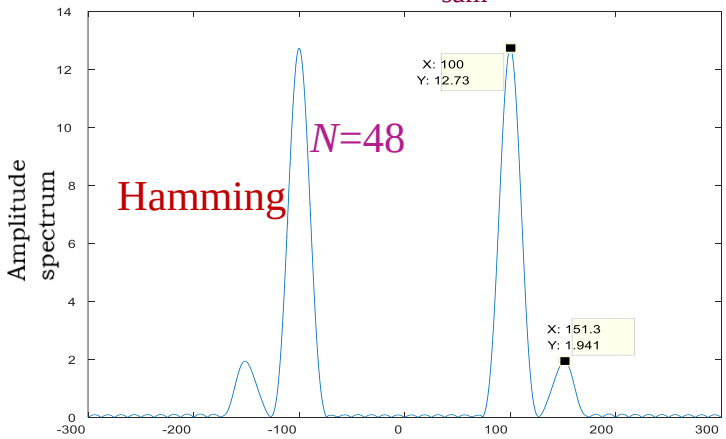
$$\Delta\Omega_w = \Delta\Omega/2$$

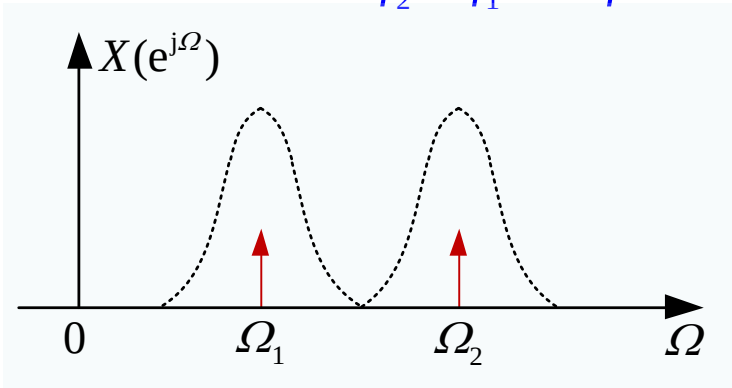
The effective width of the main lobe of a Hamming window:

$$\frac{4\pi}{N} = \frac{\pi\Delta f}{f_{sam}}$$

$$\Delta\Omega_w = 4\pi / N$$

$$N = \frac{4f_{sam}}{f_2 - f_1} = \frac{4f_{sam}}{\Delta f} = \frac{4\times 600}{50} = 48$$



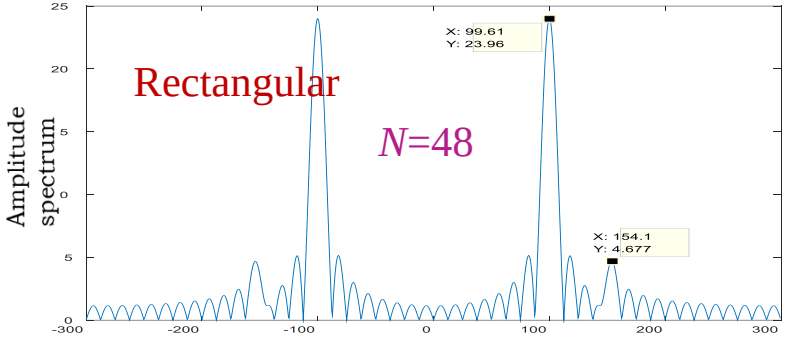


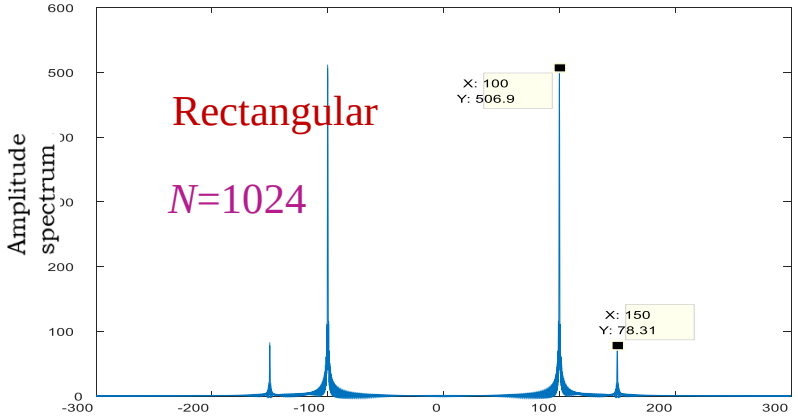
Hamming

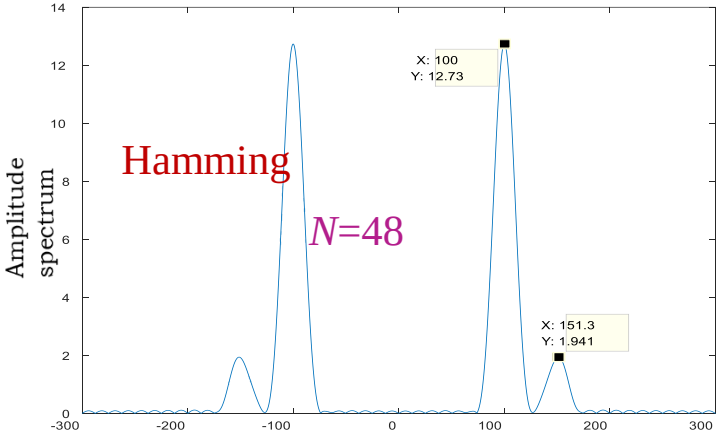
$$\Delta\Omega_w = \Delta\Omega/2$$

The ratio of the amplitude values of the two frequency points is approximately equal to 1/0.15, and both the frequency location and amplitude value estimations are correct.

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+0.15\cos(2\pi f_2 t)$, $f_1=100$Hz, $f_2=150$Hz If the signal is sampled at a sampling frequency of $f_{sam}=600$Hz , calculate the minimum number of sample points required to resolve these two spectral peaks by DFT analysis







When the amplitude values of two frequency points differ significantly, it is more appropriate to use a window with greater side lobe attenuation for windowing and truncation.

Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+0.15\cos(2\pi f_2 t)$, $f_1=100$Hz, $f_2=150$Hz. If the signal is sampled at a sampling frequency of $f_{sam}=600$Hz , calculate the spectrum analyzed by DFT.

The MATLAB program is as follows:

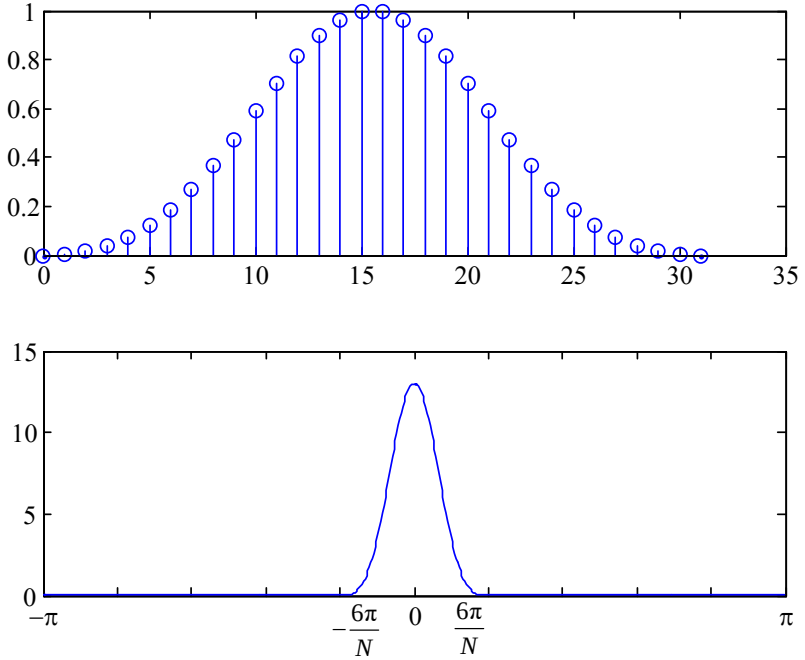
```
N=48;
L=1024;
f1=100;f2=150;
fs=600; T=1/fs;
ws=2*pi*fs;
t=(0:N-1)*T;
f=cos(2*pi*f1*t)+0.15*cos(2*pi*f2*t);
```

Using a Hamming window

```
wh=(hamming(N))';
f=f.*wh;
F=fftshift(fft(f,L));
w=(-ws/2+(0:L-1)*ws/L)/(2*pi);
plot(w,abs(F));
ylabel(' Amplitude spectrum')
xlabel(' Frequency(Hz)')
```

Leakage Phenomenon

Blackman $$w[k]=[0.42-0.5\cos(\frac{2\pi}{N-1}k)+0.08\cos(\frac{4\pi}{N-1}k)]R_N[k]$$

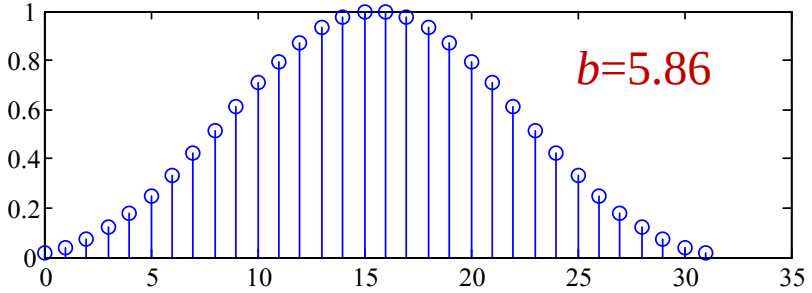


Main lobe effective width: $\Delta\Omega_w=\frac{6\pi}{N}$

Side lobe peak decay: $A\approx 57$ dB

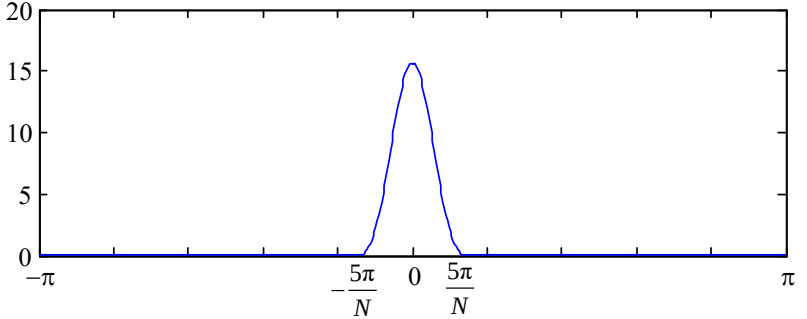
Leakage Phenomenon

Kaiser $$w_N[k]=\frac{I_0(\beta\sqrt{1-(1-2k/N)^2}}{I_0(\beta)}R_N[k]$$



Main lobe effective width: $\Delta\Omega_w=\frac{5\pi}{N}$

Side lobe peak decay: $A\approx 57$ dB



Common Characteristics of Window Functions

Type of window function	Main lobe effective width	Side lobe peak decay
Rectangle	2p /*N*	13 dB
Hanning	4p /*N*	31 dB
Hamming	4p /*N*	41 dB
Blackman	6p /*N*	57 dB
(Kaiser)*b*=5.86	5p /*N*	57 dB

Selection of Window Type

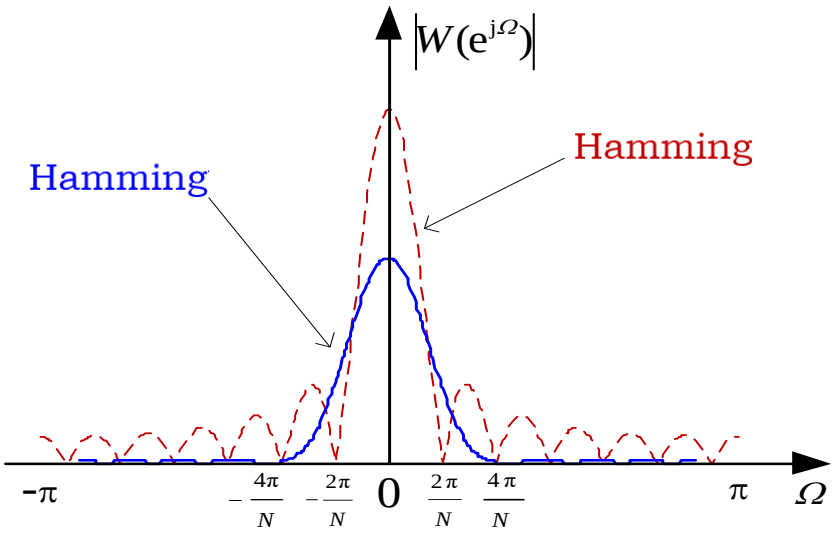
Desired window function

The narrower the main lobe width, the better

The greater the side lobe decay, the better

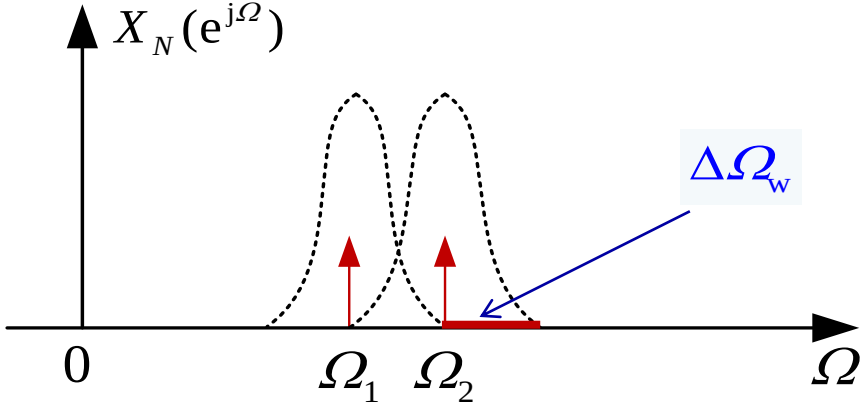
Actual window function

Narrow main lobe, small side lobe decay Wide main lobe, large side lobe decay



Choose the window function reasonably according to the actual situation

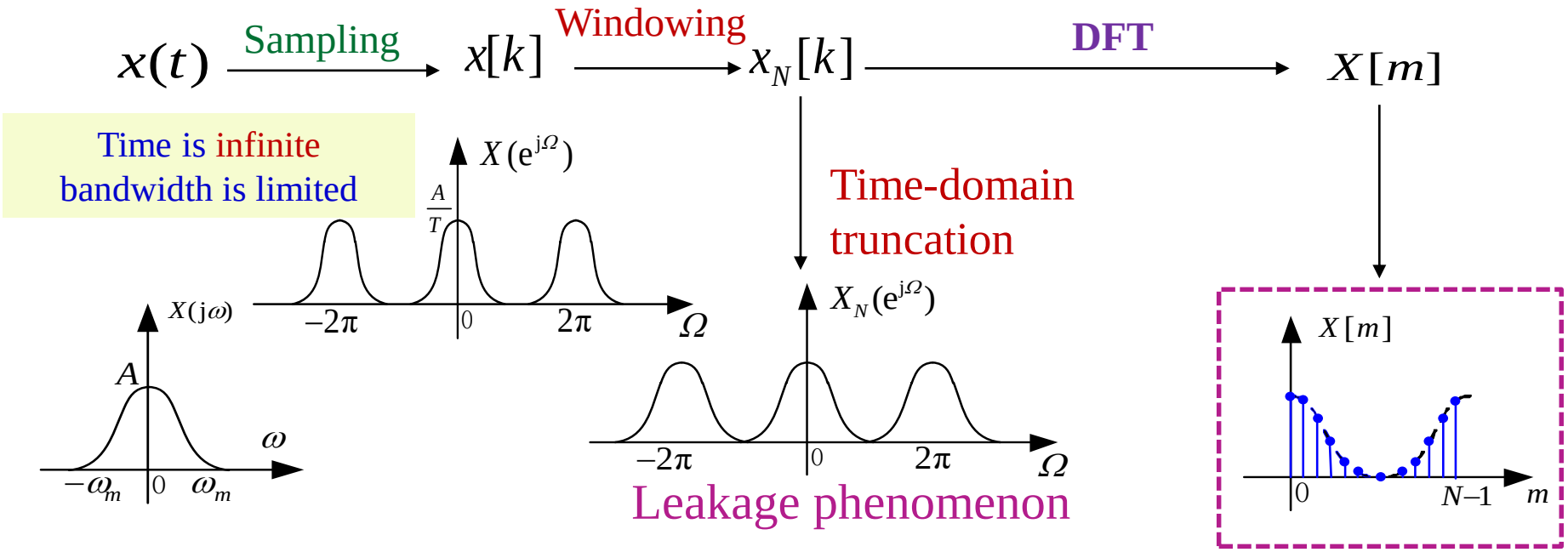
Selection of Window Type



Main lobe effective width: $\Delta\Omega_w = \frac{2\pi}{N}$ Frequency resolution: $\Delta f_c = \frac{1}{NT} = \frac{f_{sam}}{N}$

Signal acquisition time: $T_p = NT = \frac{1}{\Delta f_c}$

Fence Effect



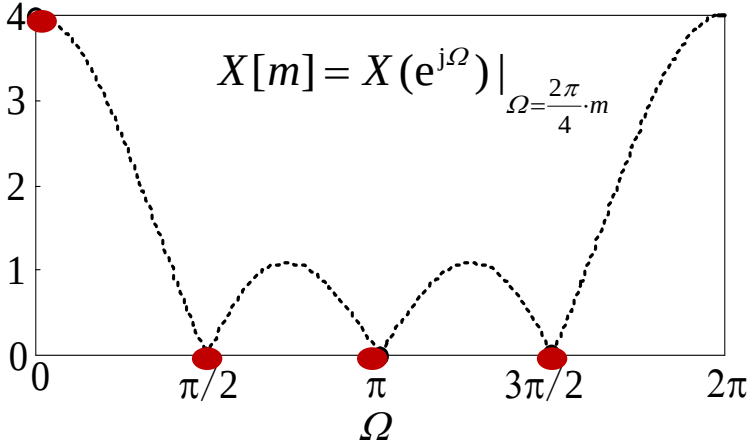
When the signal's spectrum is a continuous spectrum, using DFT to analyze the signal spectrum will produce the fence effect

Improving the fence effect: Append zeros to the sequence to increase the number of DFT points

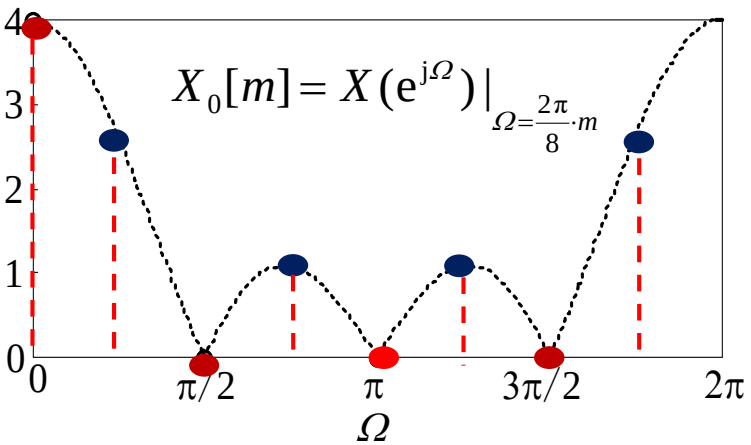
Fence Effect

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{DTFT}\{x[k]\}$$

$x[k] = \{1,1,1,1\}$



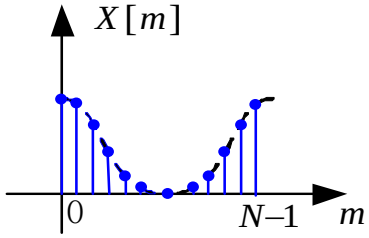
$x_0[k] = \{1,1,1,1,0,0,0,0\}$



Appending zeros to the sequence allows for the observation of more details in the spectrum

Fence Effect

$$X[m]=X_N(\mathrm{e}^{\mathrm{j}\Omega})\big|_{\Omega=\frac{2\pi}{N}m},\qquad m=0,1,\cdots,N-1$$



Spectral line spacing:

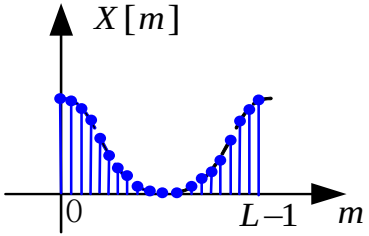
$$\Delta f_{\mathrm{d}}=\frac{f_{\mathrm{sam}}}{N}(\mathrm{Hz})$$

Zero-padding the sequence:

$$x_L[k]=\begin{cases}x_N[k] & 0\le k\le N-1\\ 0 & N\le k\le L-1\end{cases}$$

The spectral line spacing after zero-padding:

$$\Delta f_{\mathrm{d}}=\frac{f_{\mathrm{sam}}}{L}(\mathrm{Hz}),\qquad L>N$$



Example :It is known that a certain continuous signal is $x(t)=\cos(2\pi f_1 t)+\cos(2\pi f_2 t)$, $f_1=100\text{Hz}$, $f_2=120\text{Hz}$ If the signal is sampled at a sampling frequency of $f_{sam}=600\text{Hz}$ and its spectrum is analyzed by DFT. The rectangular window width $N=32$, compare the spectra obtained when the DFT point numbers are 64, 128, and 256, respectively.

Solution: $x[k]=x(t)\Big|_{t=kT}=\cos(2\pi f_1 k/f_{sam})+\cos(2\pi f_2 k/f_{sam})=\cos(\frac{\pi}{3}k)+\cos(\frac{6\pi}{15}k)$

$$x_N[k]=x[k]\cdot w_N[k]$$

$L=64$ $\quad \Delta f_d=\frac{600}{64}(\text{Hz})$

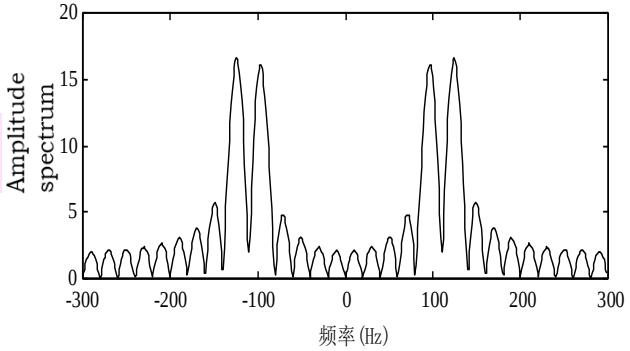
$L=128$ $\quad \Delta f_d=\frac{600}{128}(\text{Hz})$

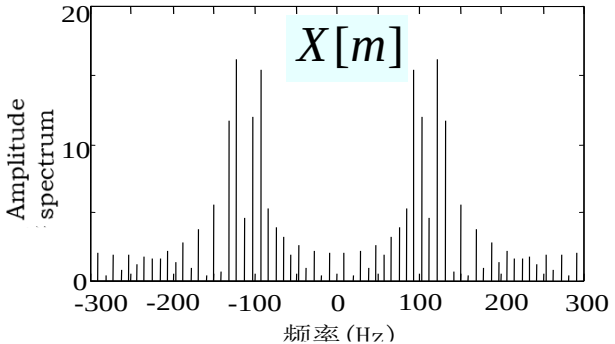
$L=256$ $\quad \Delta f_d=\frac{600}{256}(\text{Hz})$

$$x[k]=x(t)\Big|_{t=kT}=\cos(2\pi f_1 k/f_{\text{sam}})+\cos(2\pi f_2 k/f_{\text{sam}})=\cos(\frac{\pi}{3}k)+\cos(\frac{6\pi}{15}k)$$

N=30

$X_N(e^{j\Omega})$





N=30

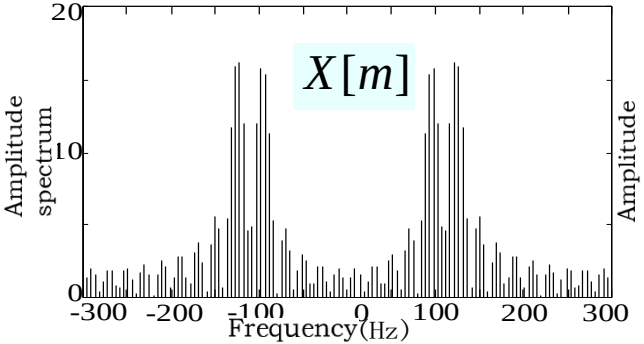
L=64,

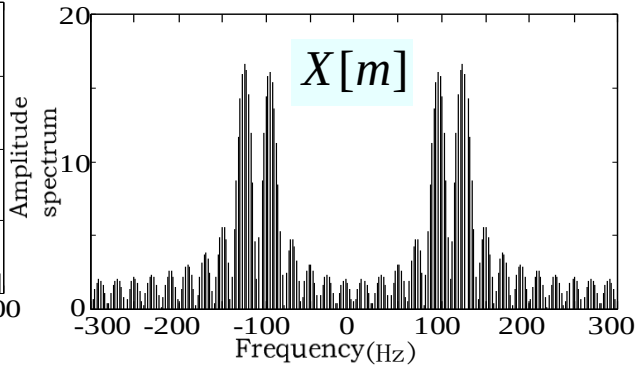
$$\Delta f_d=\frac{600}{64}$$

N=30

L=128

$$\Delta f_d=\frac{600}{128}$$





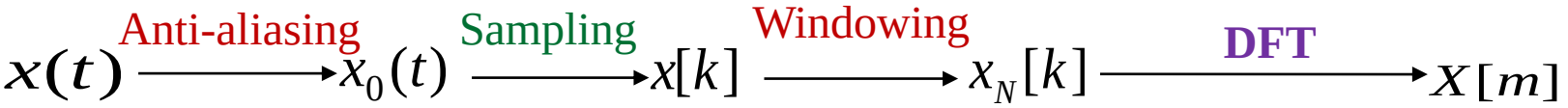
N=30

L=256

$$\Delta f_d=\frac{600}{256}$$

Frequency

Parameter Selection for Signal Spectrum Analysis by DFT



Spectral aliasing

Sampling frequency:

$$f_{\text{sam}} \geq 2 f_m$$

Spectral leakage

Acquisition time:

$$T_{\text{p}} = NT \geq \frac{1}{\Delta f_{\text{c}}}$$

Fence effect

DFT points:

$$L \geq \frac{f_{\text{sam}}}{\Delta f_{\text{d}}}$$

Example: Try to use DFT to analyze a certain continuous signal, knowing that its highest frequency is 1000 Hz, and the frequency spectrum resolution is less than 2.0 Hz, with a spectral line interval of 0.5 Hz. The number of DFT points must be an integer power of 2. Determine the following parameters: sampling interval, acquisition time, number of sampling points, and number of DFT points

Solution: $f_{\mathrm{m}} = 1000\mathrm{Hz}, \quad \Delta f_{\mathrm{c}} = 2.0\mathrm{Hz}, \quad \Delta f_{\mathrm{d}} = 0.5\mathrm{Hz}$

(1) The maximum sampling interval $T_{\max}$ is

$$T_{\max} = \frac{1}{2 f_{\mathrm{m}}} = \frac{1}{2 \times 1000} = 0.5 \times 10^{-3} (s)$$

(2) The shortest acquisition time T_{p} is

$$T_{\mathrm{p}} = NT \geq \frac{1}{\Delta f_{\mathrm{c}}} = \frac{1}{2} = 0.5 \ (\mathrm{s})$$

(3) The minimum number of sampling points N is

$$N = \frac{T_{\mathrm{p}}}{T} = \frac{0.5}{0.5 \times 10^{-3}} = 1000$$

(4) The minimum number of DFT points L is

$$L \geq \frac{f_{\mathrm{sam}}}{\Delta f_{\mathrm{d}}} = \frac{2000}{0.5} = 4000$$

Taking $L=4096=2^{12}$

DFT Analysis of the Spectrum of Non-Periodic Signals

- For an infinitely long band-limited continuous-time signal, what is the process of using DFT to analyze its spectrum? What are the errors involved? Sampling—Windowing—DFT—Establish the relationship between $X[m]$ and $X(\mathrm{j}\omega)$

  Leakage error Fence effect

- For an infinitely long discrete sequence, what is the process of using DFT to analyze its spectrum? What are the errors involved? Windowing—DFT—Establish the relationship between $X[m]$ and $X(\mathrm{e}^{\mathrm{j}\Omega})$

  Leakage error Fence effect

- For a finite-length discrete sequence, what is the process of using DFT to analyze its spectrum? What are the errors involved? DFT—Establish the relationship between $X[m]$ and $X(\mathrm{e}^{\mathrm{j}\Omega})$

  Fence effect

Utilizing DFT to analyze the signal spectrum.

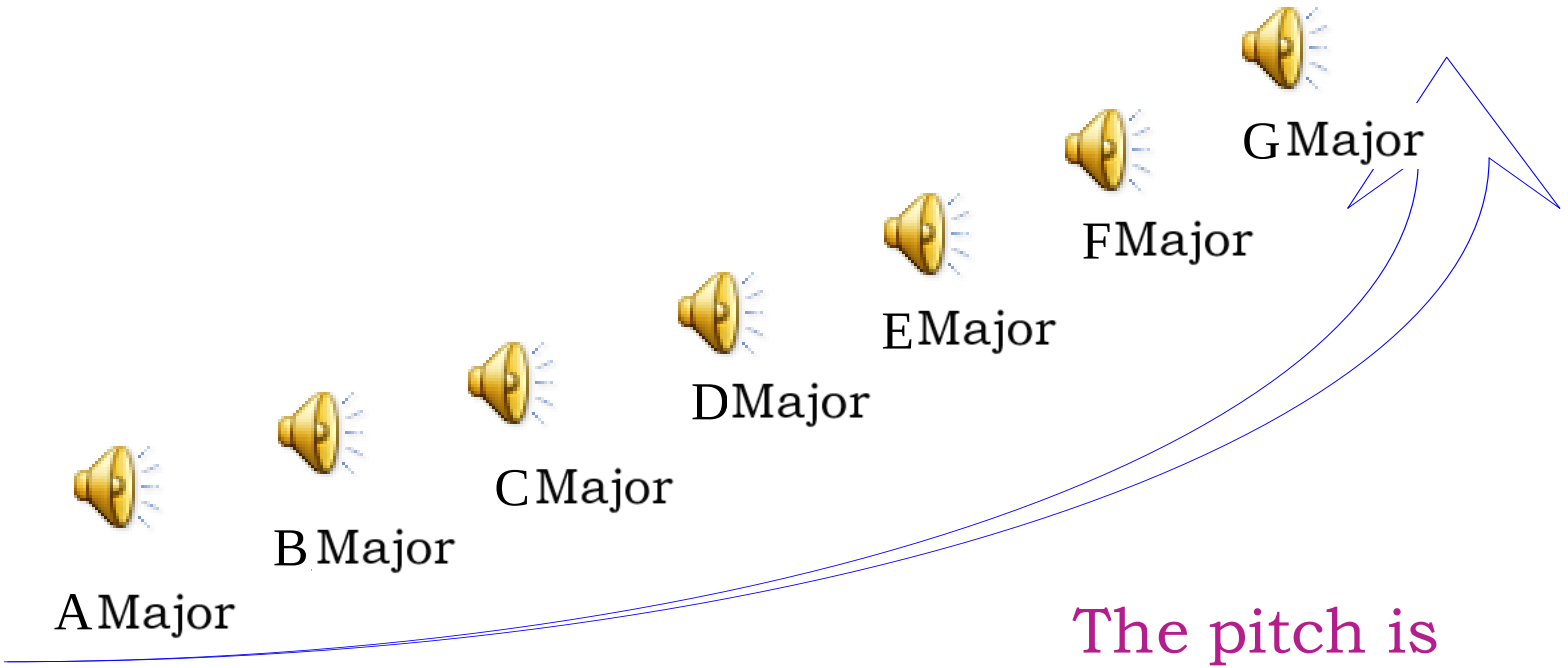
$$x(t) \xrightarrow{\text{Anti-aliasing}} x_0(t) \xrightarrow{\text{Sampling}} x[k] \xrightarrow{\text{Windowing}} x_N[k] \xrightarrow{\text{DFT}} X[m]$$

(1) Due to the windowing and truncation of the signal, spectral leakage occurs. Can it be improved by appending zeros to the truncated sequence?

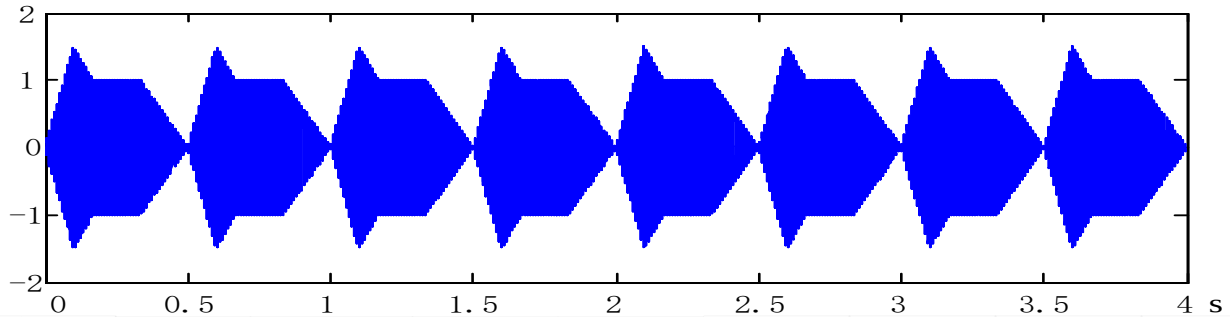
(2) Since the frequency resolution Δf_c is inversely proportional to the signal acquisition time T_p, does this mean that in engineering practice, the frequency resolution Δf_c can be easily achieved?

(3) How to reasonably choose the type of window function (Rectangular, Hanning, Hamming, Kaiser, Blackman)?

Spectral Analysis of Musical Scales



C Major Scale Time-Domain Waveform and Spectrum



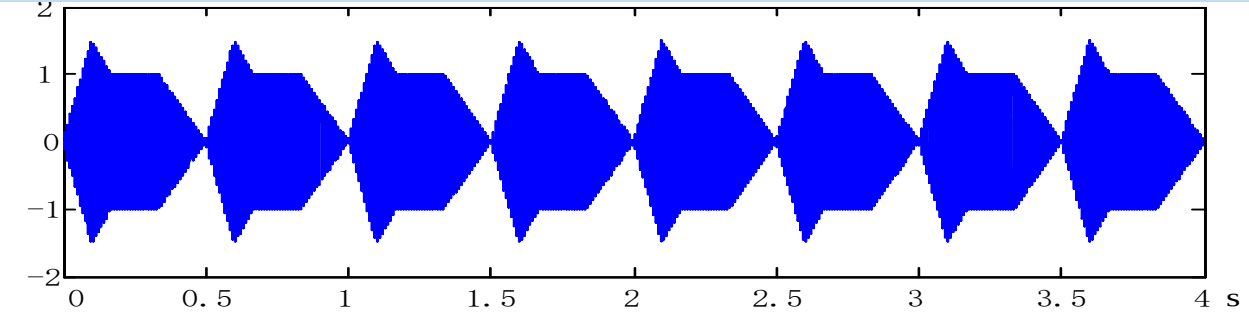
C Major

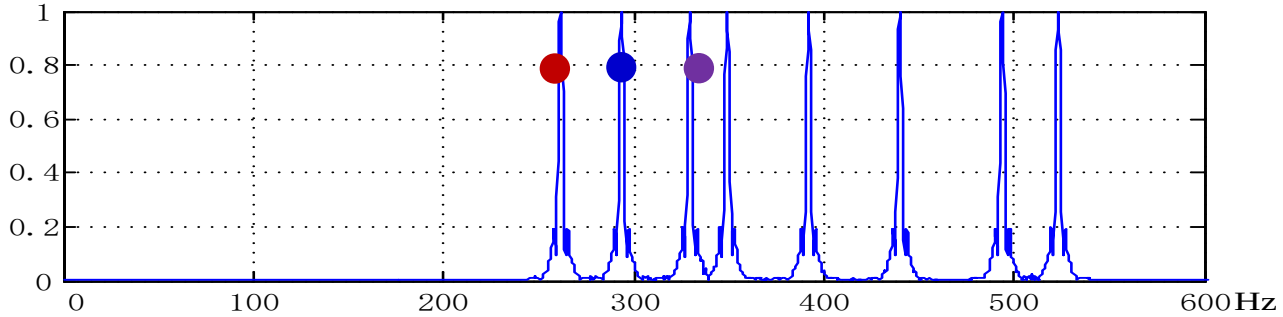
C Major

Note name	Do	Re	Mi	Fa	So	La	Si	Do^2
Frequency	262	294	330	349	392	440	494	523

```
[x,fs] = audioread('yinjie_Cdiao.wav');
N = length(x);
X = fftshift(fft(x,N));
f=-fs/2+(0:N-1)*fs/N;
plot(f,abs(X));
xlim([0,600]);
```

C Major Scale Time-Domain Waveform and Spectrum

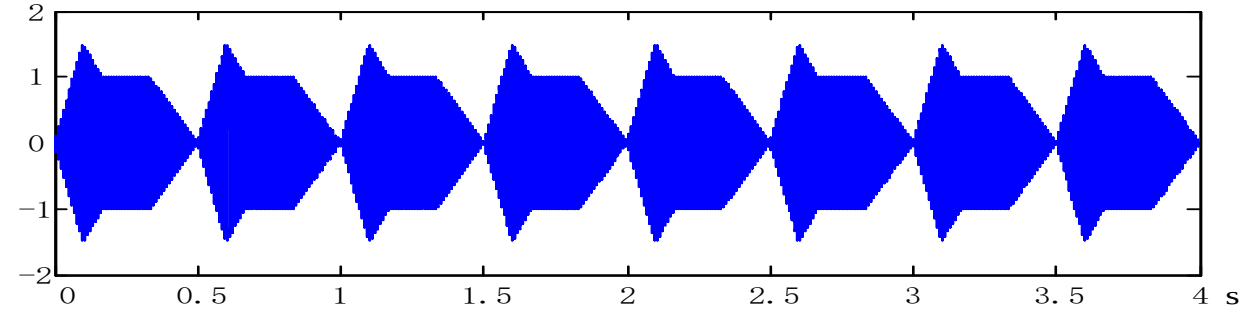


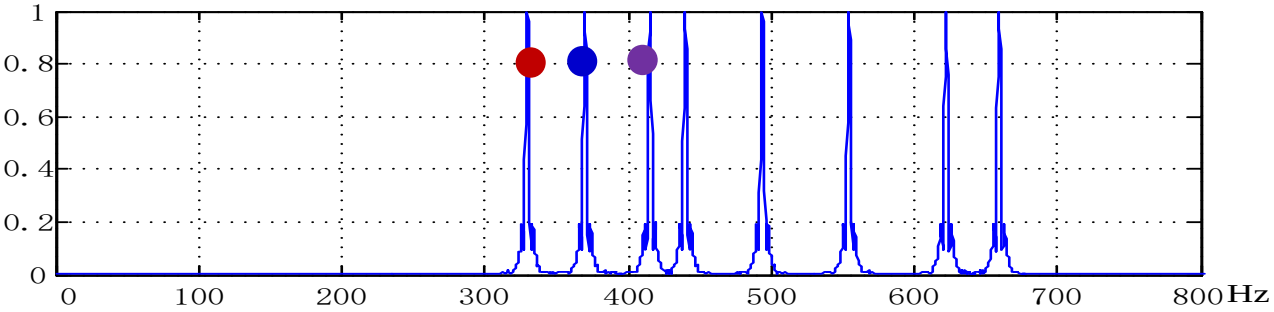


C Major

Note name	Do	Re	Mi	Fa	So	La	Si	Do^2
Frequency	262	294	330	349	392	440	494	523

E Major Scale Time-Domain Waveform and Spectrum



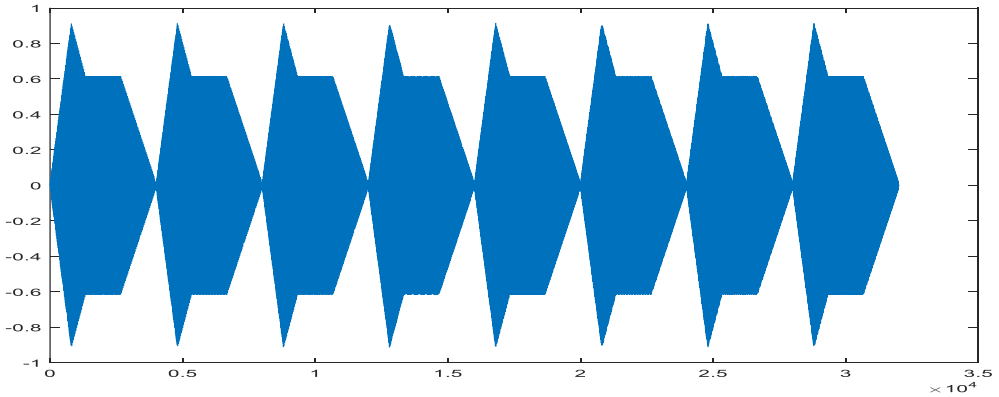


E Major

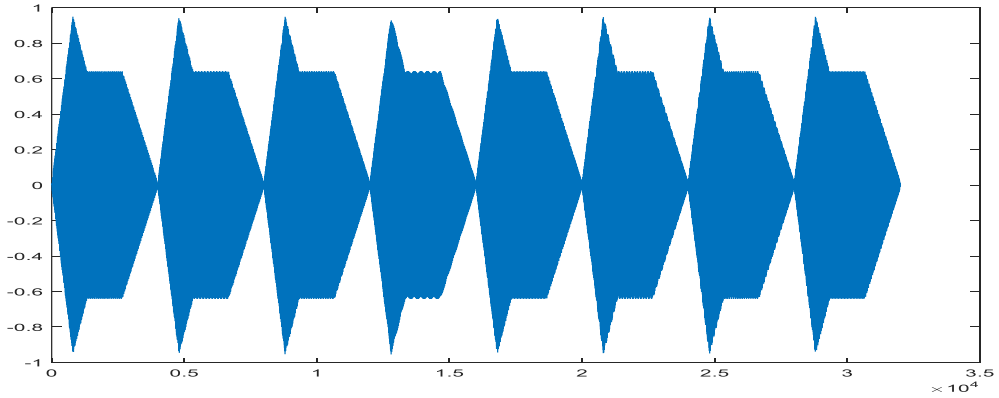
Note name	Do	Re	Mi	Fa	So	La	Si	Do^2
Frequency	330	370	415	440	494	554	622	659

Spectral Analysis of Musical Scales

Without harmonics



Containing harmonics



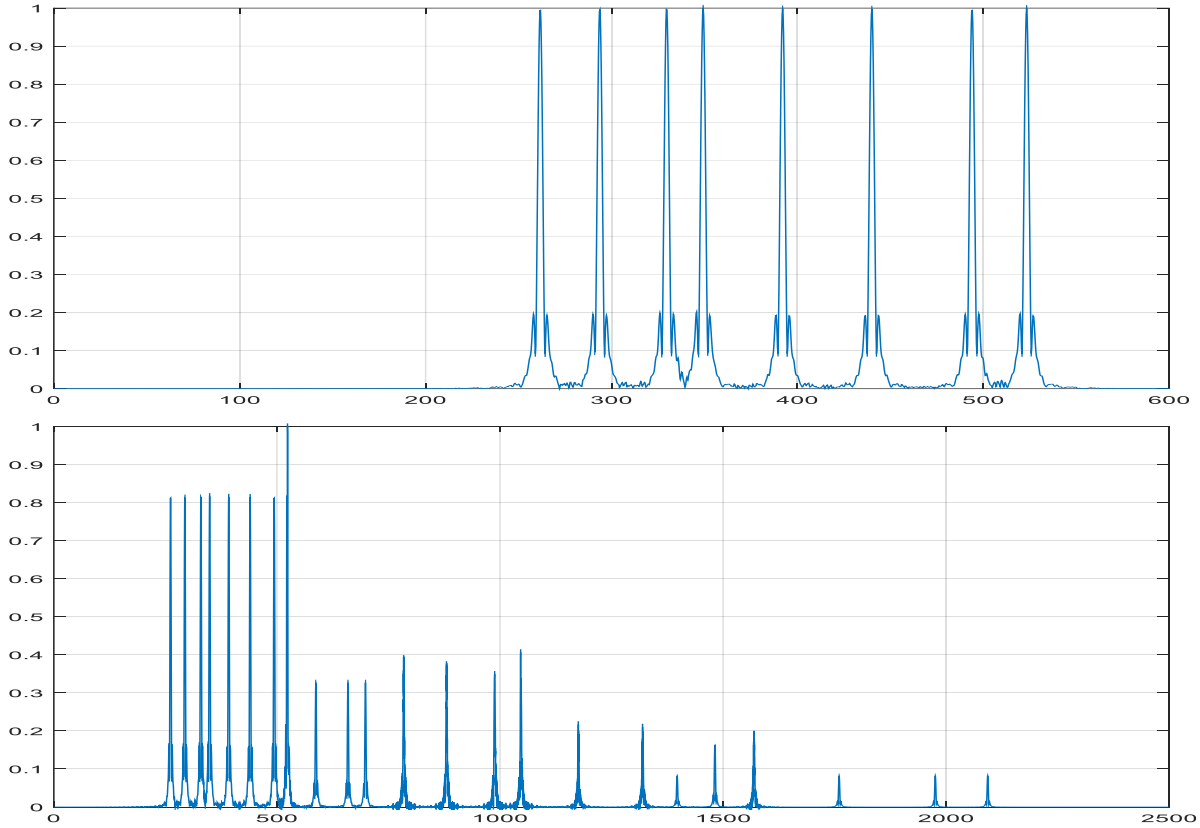
The timbre of the two is different

Spectral Analysis of Musical Scales

Without harmonics

Containing harmonics

The timbre of the two is different

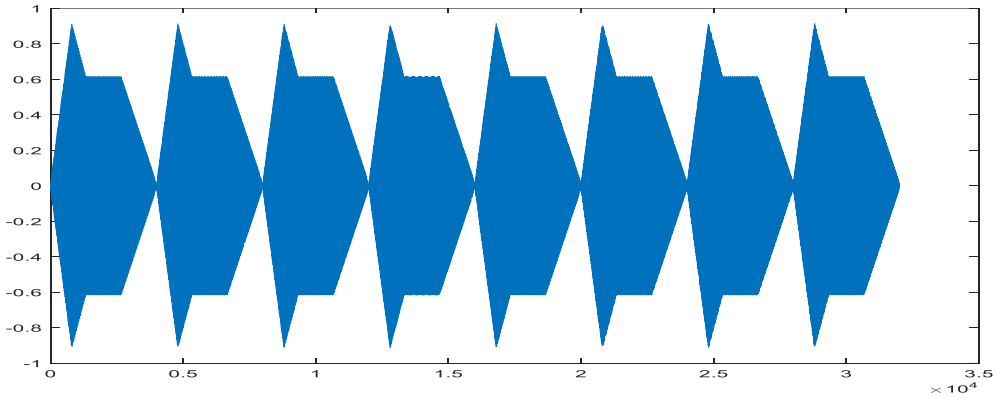


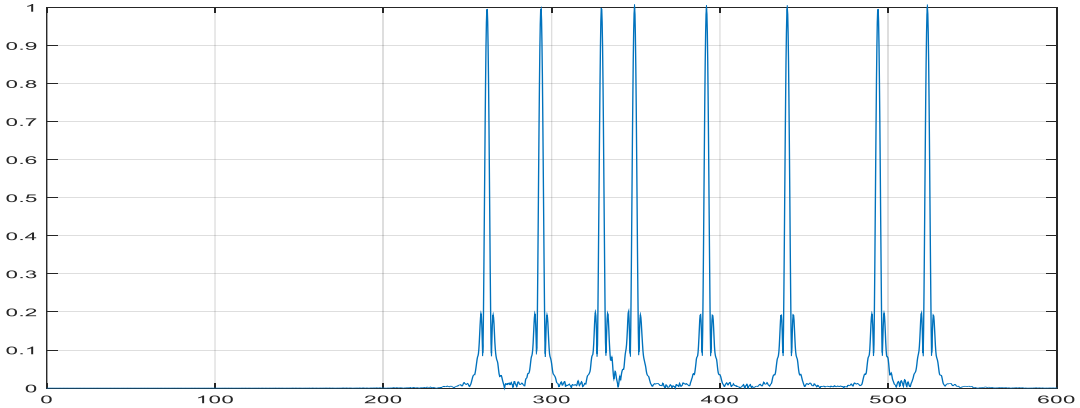
Spectral Analysis of Musical Scales



Without harmonics

A scale corresponds to a single frequency point



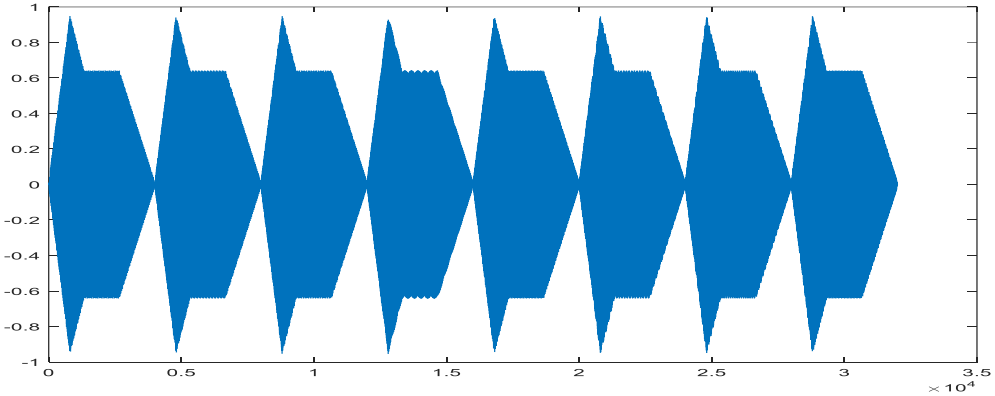


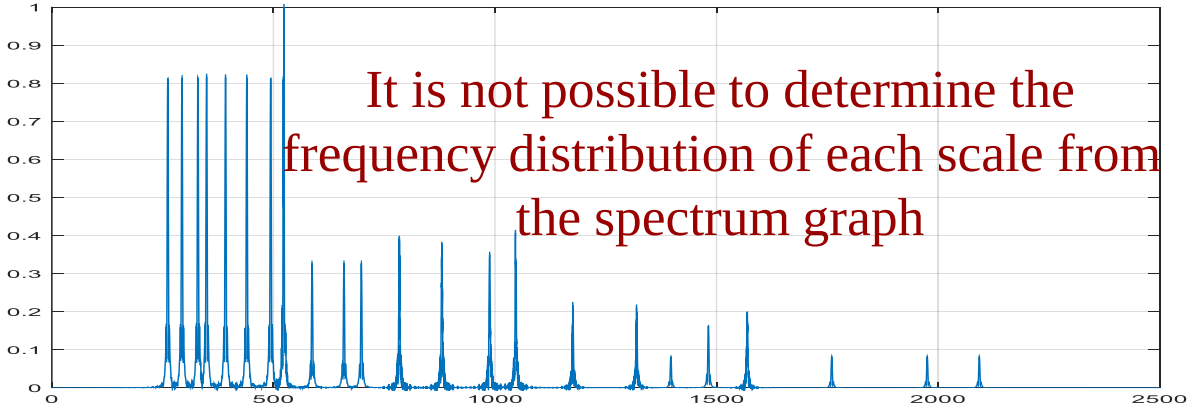
Spectral Analysis of Musical Scales



Containing harmonics

A scale corresponds to multiple frequency points

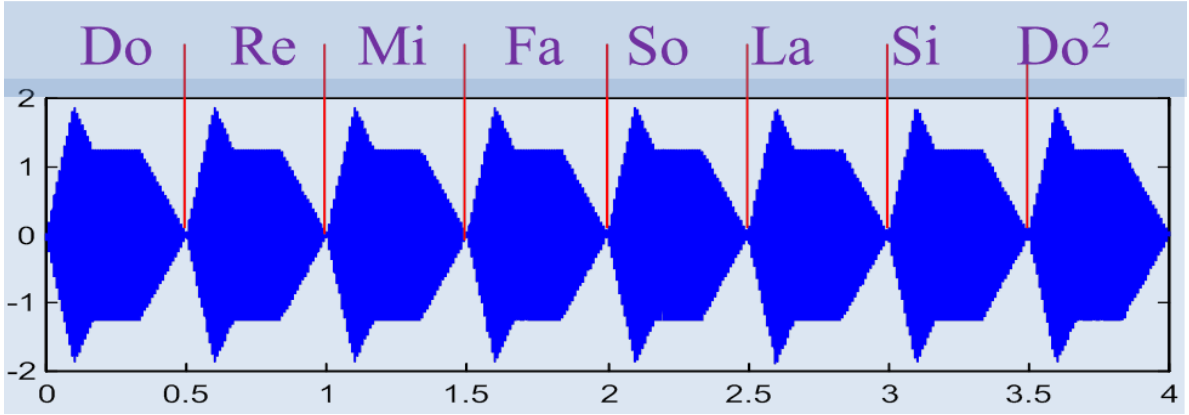




It is not possible to determine the frequency distribution of each scale from the spectrum graph

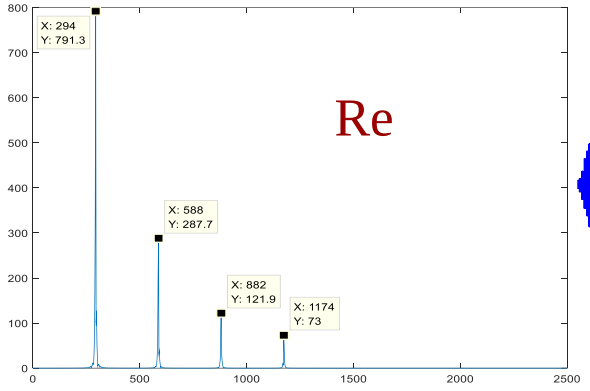
Spectral Analysis of Musical Scales

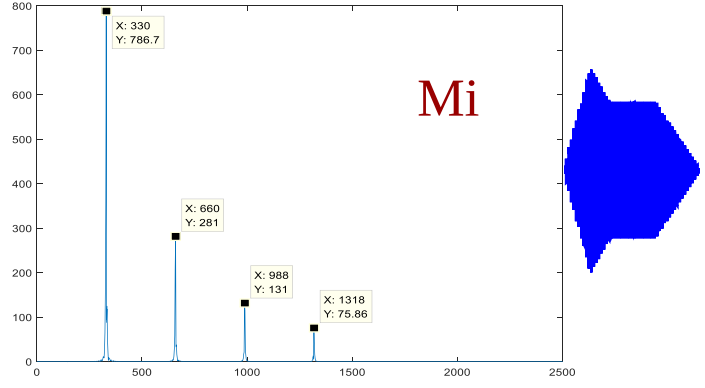
Based on the characteristics of the time-domain waveform, segment out each scale——Windowing and truncation



Obtain the frequency composition of each scale

Spectral Analysis of Musical Scales





Each pitch is composed of sinusoidal components with different fundamental frequencies

C Major scale frequency spectrum

Scale	Fundamental frequency	Harmonic distribution(Frequency Hz/Amplitude)		
Do	262	524/ 0.38	786/0.18	1048/0.1
Re	294	588/ 0.38	882/0.18	1176/0.1
Mi	330	660/ 0.38	990/0.18	1320/0.1

Applications of DFT Analysis in Signal Spectrum with Examples

Identify the song instantly by listening

原 Browse: 29756 Update: 2014-03-06 15:01



在商场超市或咖啡厅等陌生场所听到一首很带感的歌曲，你无从打听这是首什么歌而又很想知道，怎么办？

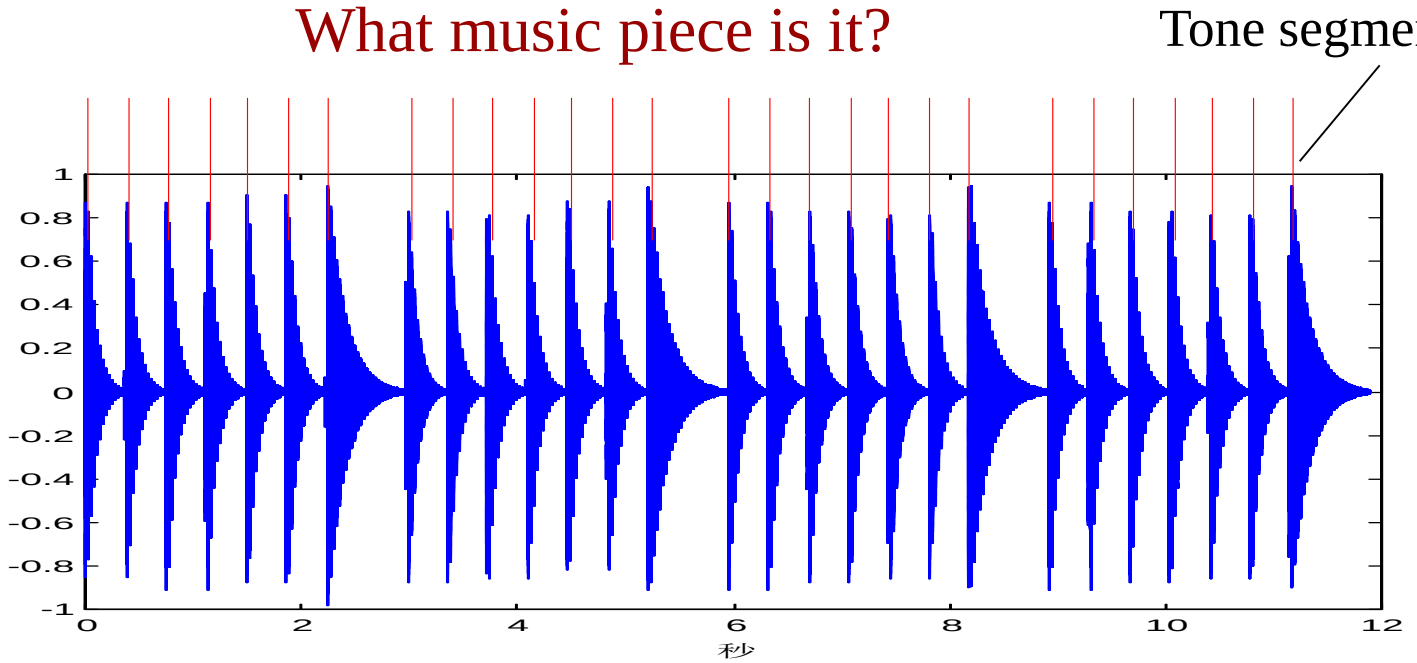
目前智能手机上已经有相应的软件能解决这一问题了，只要把软件打开，把手机对着音源，不出几秒就能查找出歌曲，也就是**听音识曲**或**听歌识曲**。

Mobile App



Music notation for digital music

What music piece is it?

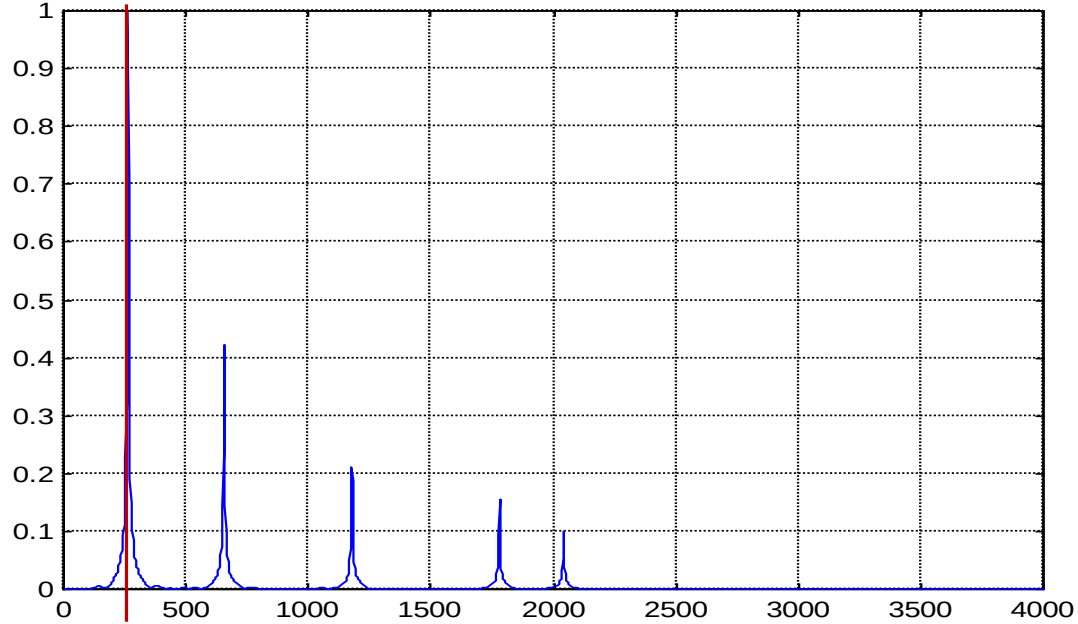


The time-domain waveform of the music piece

Music notation for digital music

Analyze the frequency spectrum program of the first musical note

```
[m,fs]=wavread('Music piece'
L=length(m);
N=L/32;
m1=m(1:N);
fm1=fft(m1);
f=(0:N/2-1)*fs/N;
plot(f,abs(fm1(1:N/2))/max(a
grid on;
axis([0,4000,0,1])
```



The fundamental frequency is approximately260Hz

→ *C* Major 1(Do)

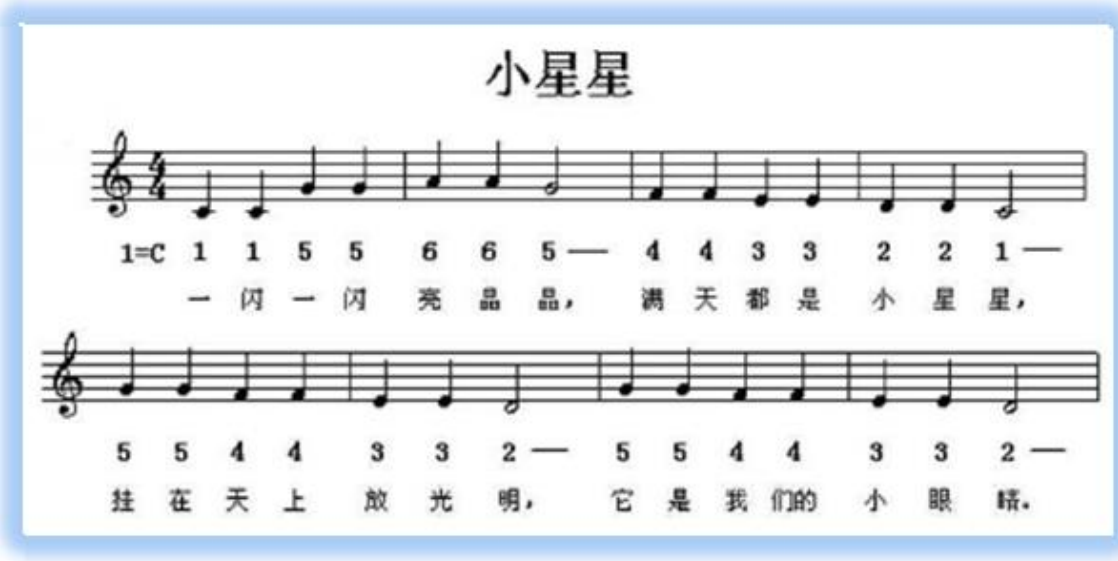
Music notation for digital music

Analyze the first musical note's frequency spectrum program

```
[m,fs]=wavread('Music piece ');
L=length(m);
N=L/32;
m1=m(1:N);
fm1=fft(m1);
f=(0:N/2-1)*fs/N;
plot(f,abs(fm1(1:N/2))/max(ab
grid on;
axis([0,4000,0,1])
```

By the same method, the frequency spectra of various musical notes can be analyzed

Identify the name of the music piece.



Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.